# Lecture 2:
# Image Classification

# Waitlisted Students

- We'll continue sending overrides in waitlist order as seats open
- If you get one, **enroll right now** – override will expire in 24h
- If you received an override and it expires, **you will not get another**

# Office Hours

Office hours start this week; check Google Calendar for details

Fill out the following form about your office hour time preferences:
https://forms.gle/Qh41oXuzjrTxFEVk9

# Piazza

Reminder: Piazza is our main source of communication this semester

Right now (enrolled students) < (enrolled students on Piazza)

Go enroll on Piazza if you haven't already

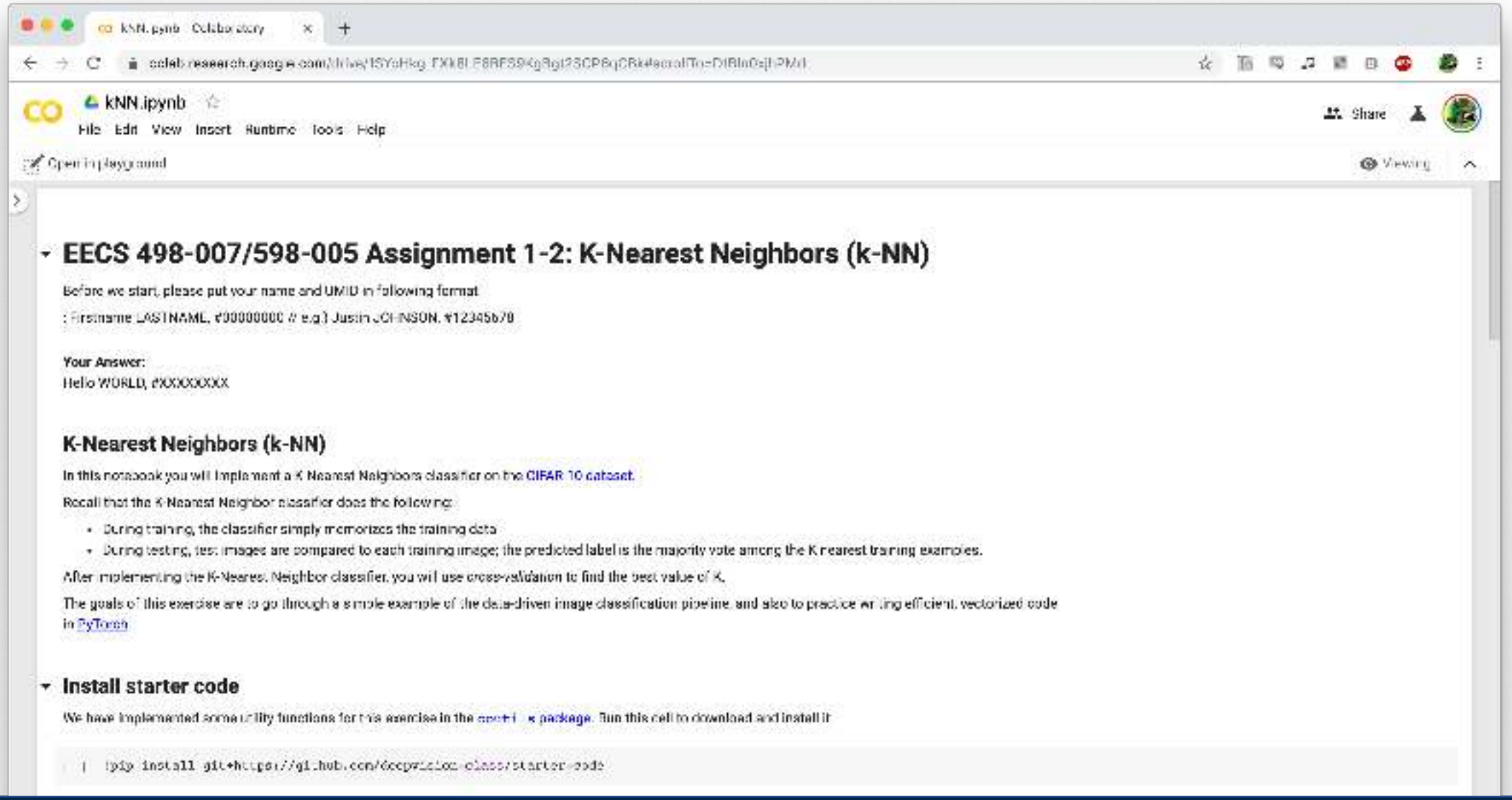
# Assignment 1 Released

- https://web.eecs.umich.edu/~justincj/teaching/eecs498/WI2022/assignment1.html
- Uses Python, PyTorch, and Google Colab
- Introduction to PyTorch Tensors
- K-Nearest Neighbor classification
- Two *challenge questions* worth 2% each
- Due **Friday January 14, 11:59pm ET**

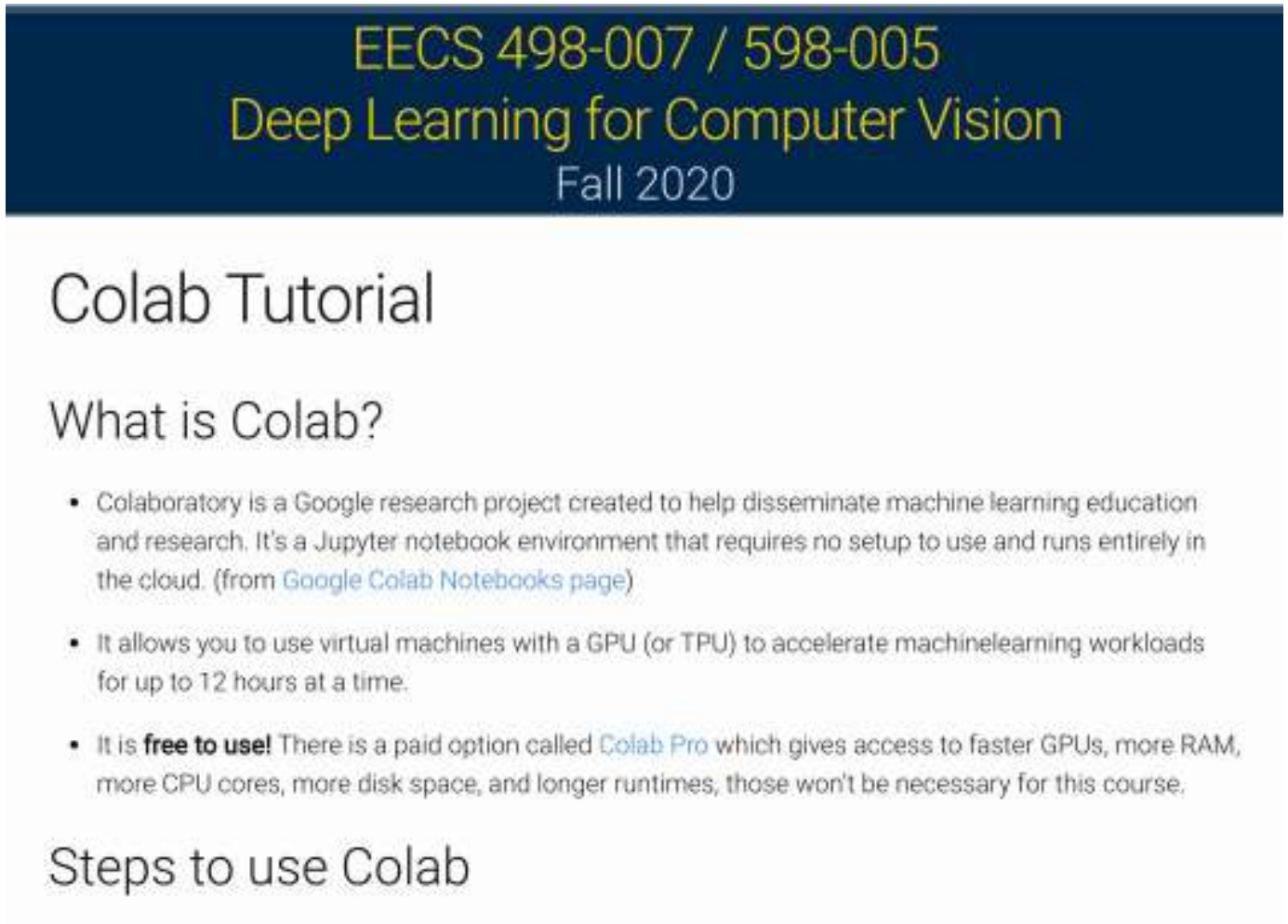
# Assignment 1 Released

- https://web.eecs.umich.edu/~justincj/teaching/eecs498/WI2022/assignment1.html
- Make sure you download the **WI2022** version of the assignment!
- We released a slightly updated assignment today with minor bugfixes; see Piazza: https://piazza.com/class/kxtai72amx34p0?cid=46
- Only **enrolled students** can submit the assignment, but if you enroll late we will give you extra days to submit A1

# Google Colab: Cloud Computing in the Browser



# Google Colab: Cloud Computing in the Browser



EECS 498-007 / 598-005
Deep Learning for Computer Vision
Fall 2020

## Colab Tutorial

### What is Colab?

- Colaboratory is a Google research project created to help disseminate machine learning education and research. It's a Jupyter notebook environment that requires no setup to use and runs entirely in the cloud. (from Google Colab Notebooks page)
- It allows you to use virtual machines with a GPU (or TPU) to accelerate machinelearning workloads for up to 12 hours at a time.
- It is **free to use!** There is a paid option called Colab Pro which gives access to faster GPUs, more RAM, more CPU cores, more disk space, and longer runtimes, those won't be necessary for this course.

### Steps to use Colab

We've written a Colab tutorial: https://web.eecs.umich.edu/~justincj/teaching/eecs498/WI2022/colab.html

# Lecture 2:
# Image Classification

# **Image Classification**: A core computer vision task

**Input**: image

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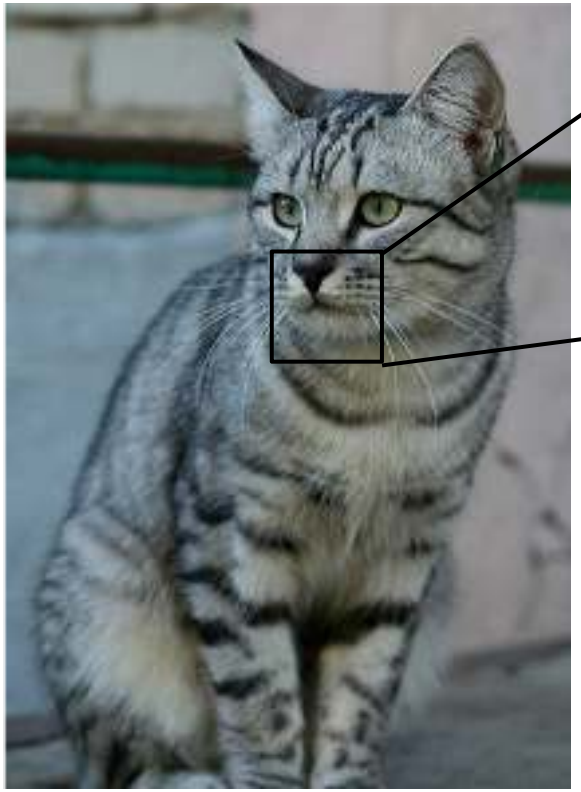
**Output**: Assign image to one of a fixed set of categories

cat
bird
deer
dog
truck

# **Problem**: Semantic Gap



This image by Nikita is licensed under CC-BY 2.0

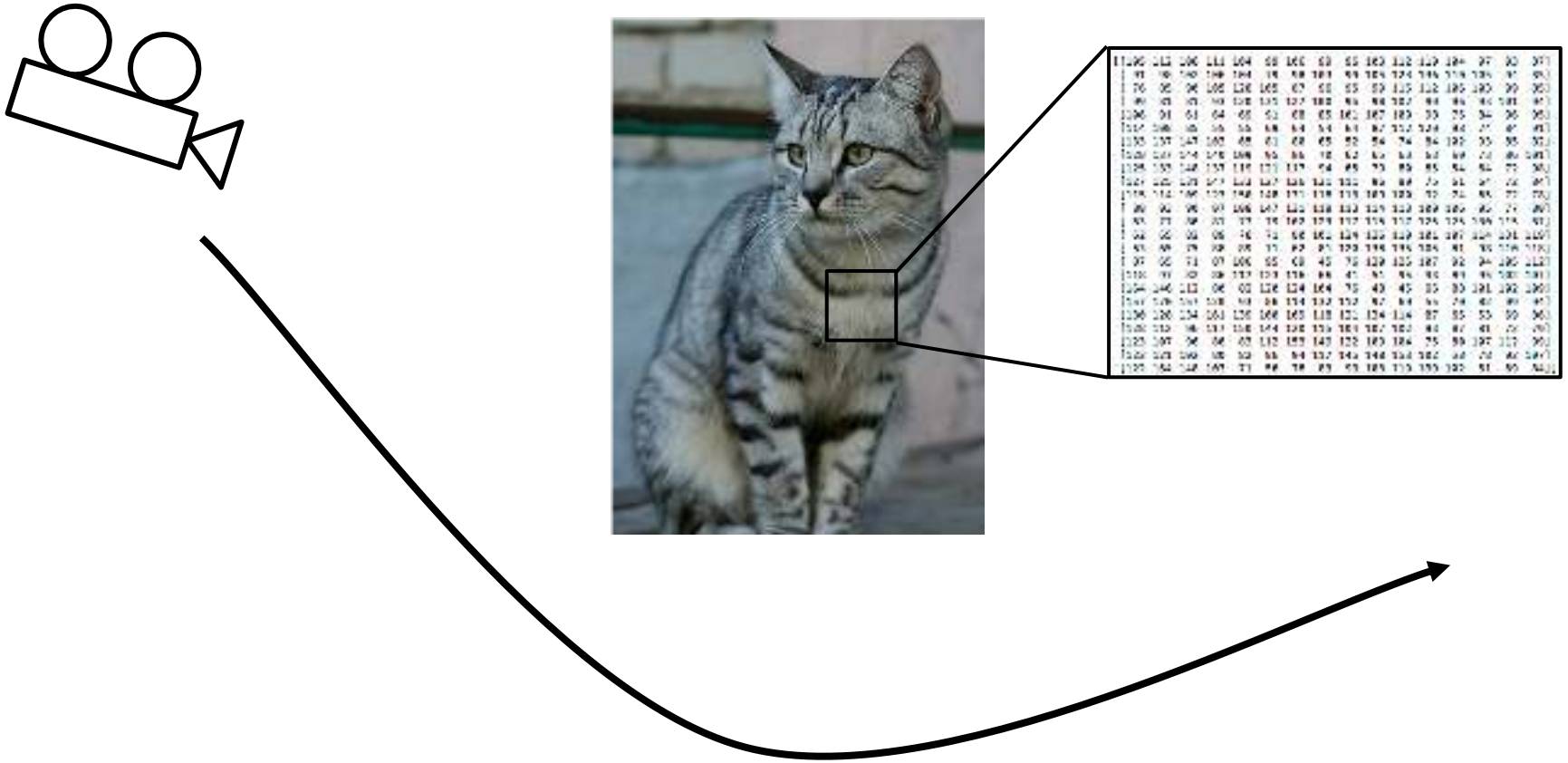
```
[[105 112 108 111 104  99 106  99  96 103 112 119 104  97  93  87]
 [ 91  98 102 106 104  79  98 103  99 105 123 136 110 105  94  85]
 [ 76  85  98 105 128 105  87  96  95  99 115 112 100 103  99  85]
 [ 99  81  81  93 120 131 127 108  95  98 102  99  96  93 101  94]
 [106  91  61  64  60  91  88  85 101 107 109  98  75  84  96  95]
 [114 108  85  55  55  60  64  54  64  87 112 120  98  74  84  91]
 [133 137 147 103  65  81  80  65  52  54  74  84 102  93  85  82]
 [128 137 144 140 109  95  86  70  52  65  63  63  60  73  86 101]
 [125 133 140 137 119 121 117  94  65  79  80  65  54  64  72  98]
 [127 125 131 147 133 127 126 131 111  96  89  75  61  64  72  84]
 [115 114 104 123 150 148 131 118 113 109 108  92  74  65  72  78]
 [ 89  93  98  97 108 147 131 118 113 114 113 109 106  95  77  88]
 [ 63  77  86  81  77  70 102 123 117 115 117 125 125 130 115  87]
 [ 62  65  82  89  78  71  80 101 124 126 119 101 107 114 131 119]
 [ 63  65  75  88  89  71  62  81 120 138 135 105  81  98 110 118]
 [ 87  65  71  87 105  95  69  45  76 130 126 107  92  94 105 112]
 [118  97  82  86 117 123 116  66  41  51  95  93  89  95 102 107]
 [164 146 112  88  82 120 124 104  75  48  45  66  88 101 102 109]
 [157 170 157 120  93  86 114 132 112  97  69  55  70  82  99  94]
 [130 128 134 161 139 100 109 118 121 134 114  87  65  53  69  86]
 [128 112  96 117 150 144 120 115 104 107 102  93  87  81  72  79]
 [123 107  96  86  83 112 153 149 122 109 104  75  80 107 112  99]
 [122 121 102  80  82  86  94 117 145 148 153 102  58  78  92 107]
 [122 104 148 103  71  56  78  83  93 103 119 139 102  61  69  84]]
```

What the computer sees

An image is just a big grid of numbers between [0, 255]:

e.g. 800 x 600 x 3
(3 channels RGB)

# **Challenges**: Viewpoint Variation



All pixels change when the camera moves!

# Challenges: Intraclass Variation

# Challenges: Fine-Grained Categories

Maine Coon

This image is free for for use under the Pixabay License

Ragdoll

This image is CC0 public domain

American Shorthair

This image is CC0 public domain

# Challenges: Background Clutter

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This image is CC0 1.0 public domain

# Challenges: Illumination Changes

This image is CC0 1.0 public domain

This image is CC0 1.0 public domain

This image is CC0 1.0 public domain

This image is CC0 1.0 public domain

# Challenges: Deformation

This image by Umberto Salvagnin is licensed under CC-BY 2.0

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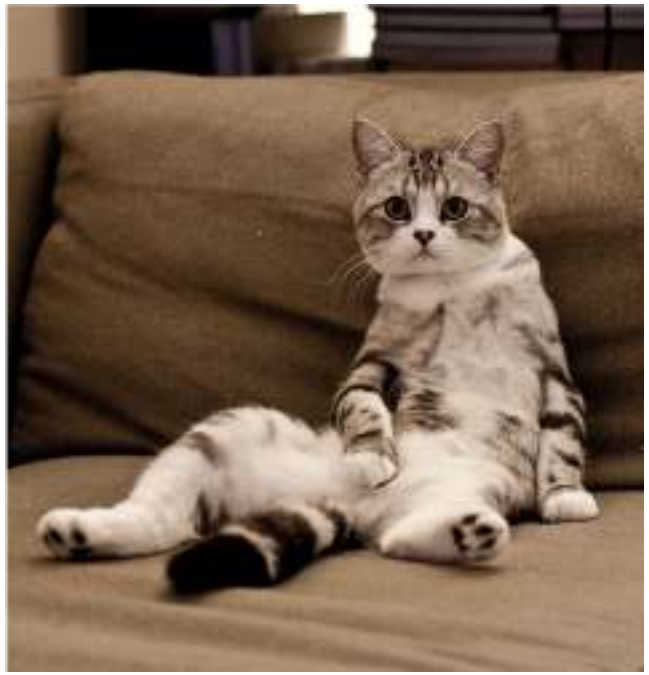
This image by sare bear is licensed under CC-BY 2.0



This image by Tom Thai is licensed under CC-BY 2.0

# Challenges: Occlusion

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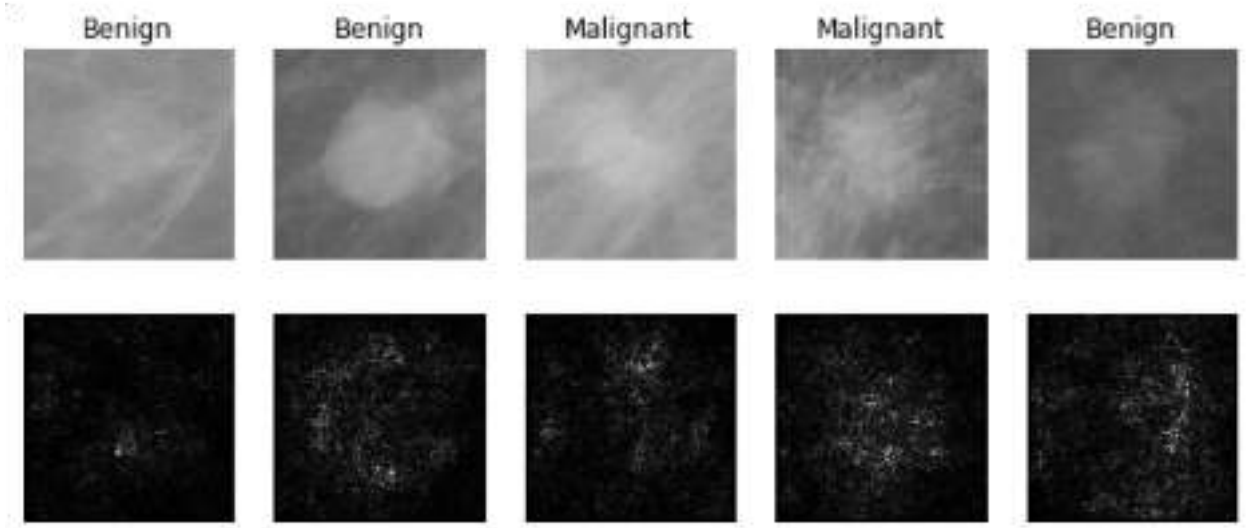
This image is CC0 1.0 public domain

This image by jonsson is licensed under CC-BY 2.0

# Image Classification: Very Useful!

Medical Imaging




Levy et al, 2016 Figure reproduced with permission

Galaxy Classification

Dieleman et al, 2014

From left to right: public domain by NASA, usage permitted by ESA/Hubble, public domain by NASA, and public domain.

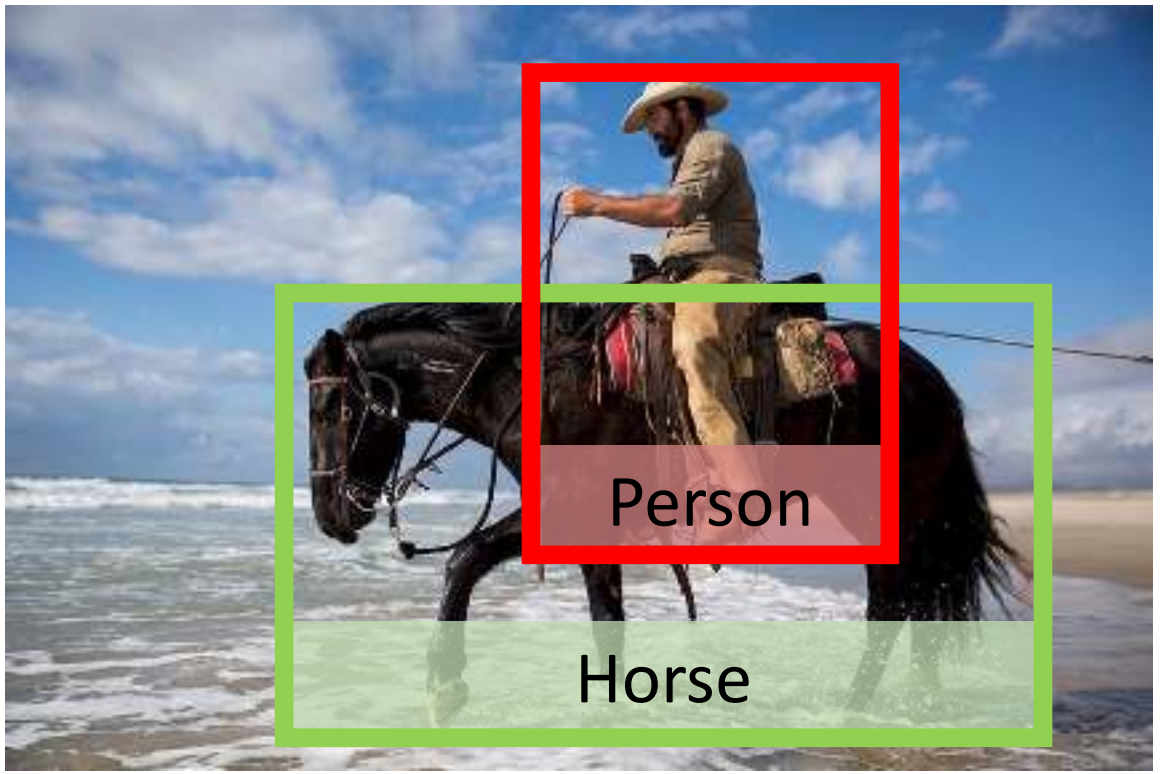
Whale recognition

Kaggle Challenge

This image by Christin Khan is in the public domain and originally came from the U.S. NOAA.

# Image Classification: Building Block for other tasks!

## Example: Object Detection



# Image Classification: Building Block for other tasks!

## Example: Object Detection



Background

Horse

Person

Car

Truck

# Image Classification: Building Block for other tasks!

## Example: Object Detection



→

Background
Horse
Person
Car
Truck

# Image Classification: Building Block for other tasks!

## Example: Image Captioning



riding
cat
horse
**man**
when
…
<STOP>

What word to say next?

Caption:
Man

# Image Classification: Building Block for other tasks!

## Example: Image Captioning

This image is free to use under the Pexels license

riding
cat
horse
man
when
…
<STOP>

What word to say next?

Caption:
Man riding

# Image Classification: Building Block for other tasks!

## Example: Image Captioning



riding
cat
**horse**
man
when
…
<STOP>

What word
to say next?

Caption:
Man riding horse

# Image Classification: Building Block for other tasks!

## Example: Image Captioning



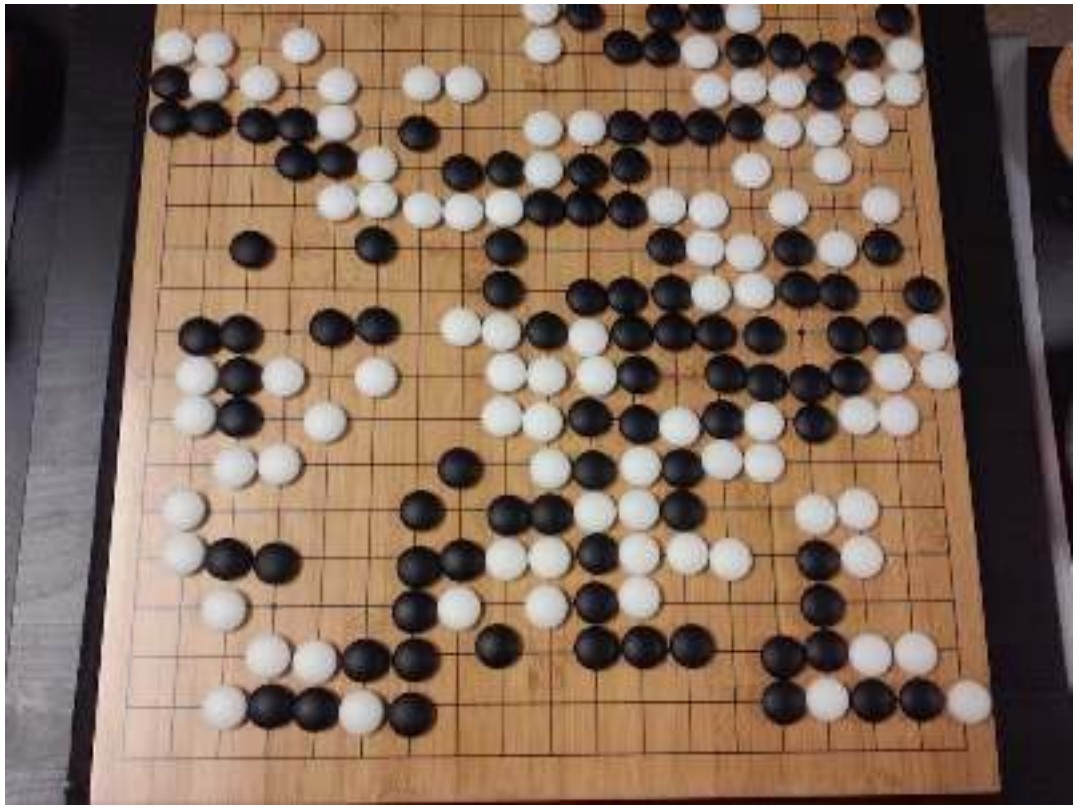
riding
cat
horse
man
when
…
<STOP>

What word
to say next?

Caption:
Man riding horse

# Image Classification: Building Block for other tasks!

Example: Playing Go



This image is CC0 public domain

(1, 1)
(1, 2)
…
(1, 19)
…
(19, 19)

Where to play next?

# An Image Classifier

```
def classify_image(image):
  # Some magic here?
  return class_label
```

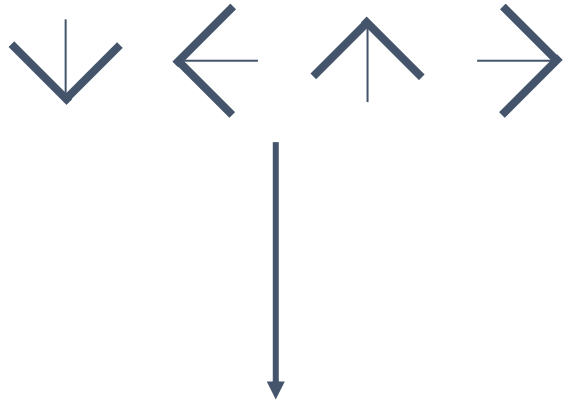
Unlike e.g. sorting a list of numbers,

**no obvious way** to hard-code the algorithm for recognizing a cat, or other classes.

# You could try …

Find edges

Find corners



?

John Canny, “A Computational Approach to Edge Detection”, IEEE TPAMI 1986

# Machine Learning: Data-Driven Approach

1. Collect a dataset of images and labels
2. Use Machine Learning to train a classifier
3. Evaluate the classifier on new images

```
def train(images, labels):
  # Machine learning!
  return model
```

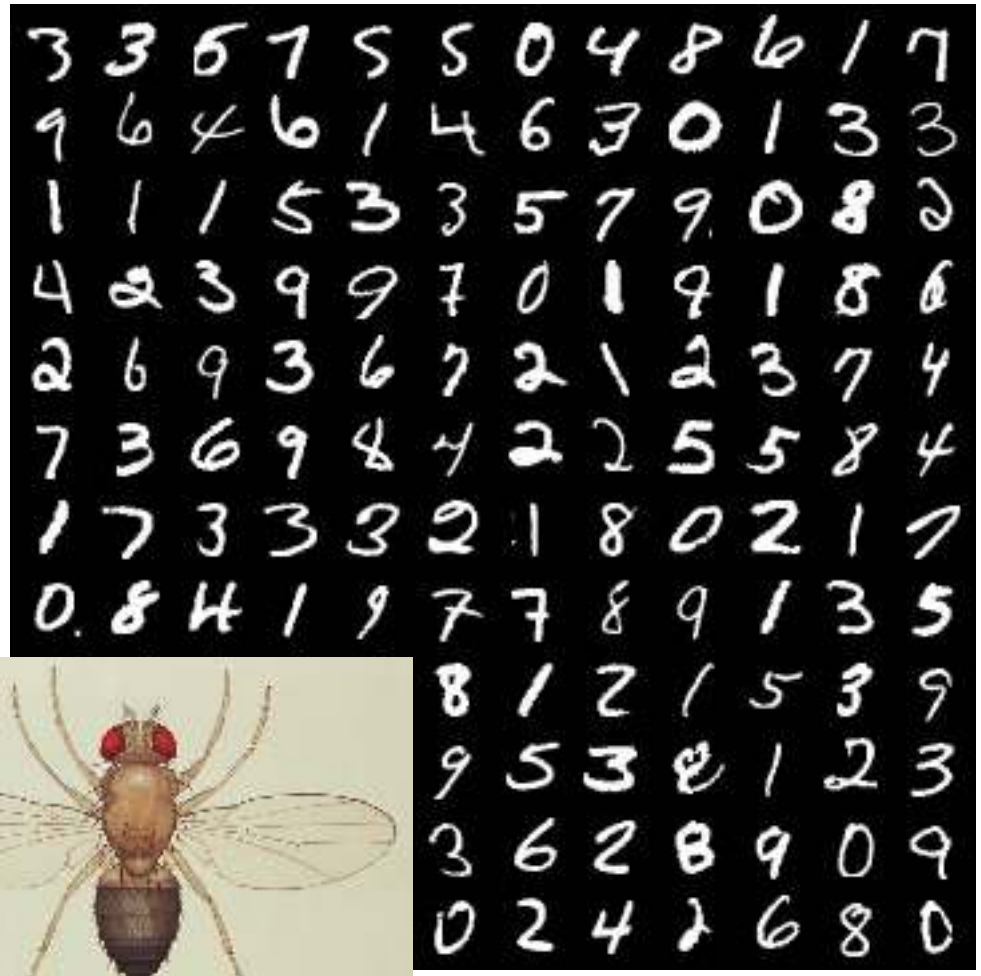
```
def predict(model, test_images):
  # Use model to predict labels
  return test_labels
```

**Example training set**

# Image Classification Datasets: MNIST

**10 classes**: Digits 0 to 9
**28x28** grayscale images
**50k** training images
**10k** test images

# Image Classification Datasets: MNIST



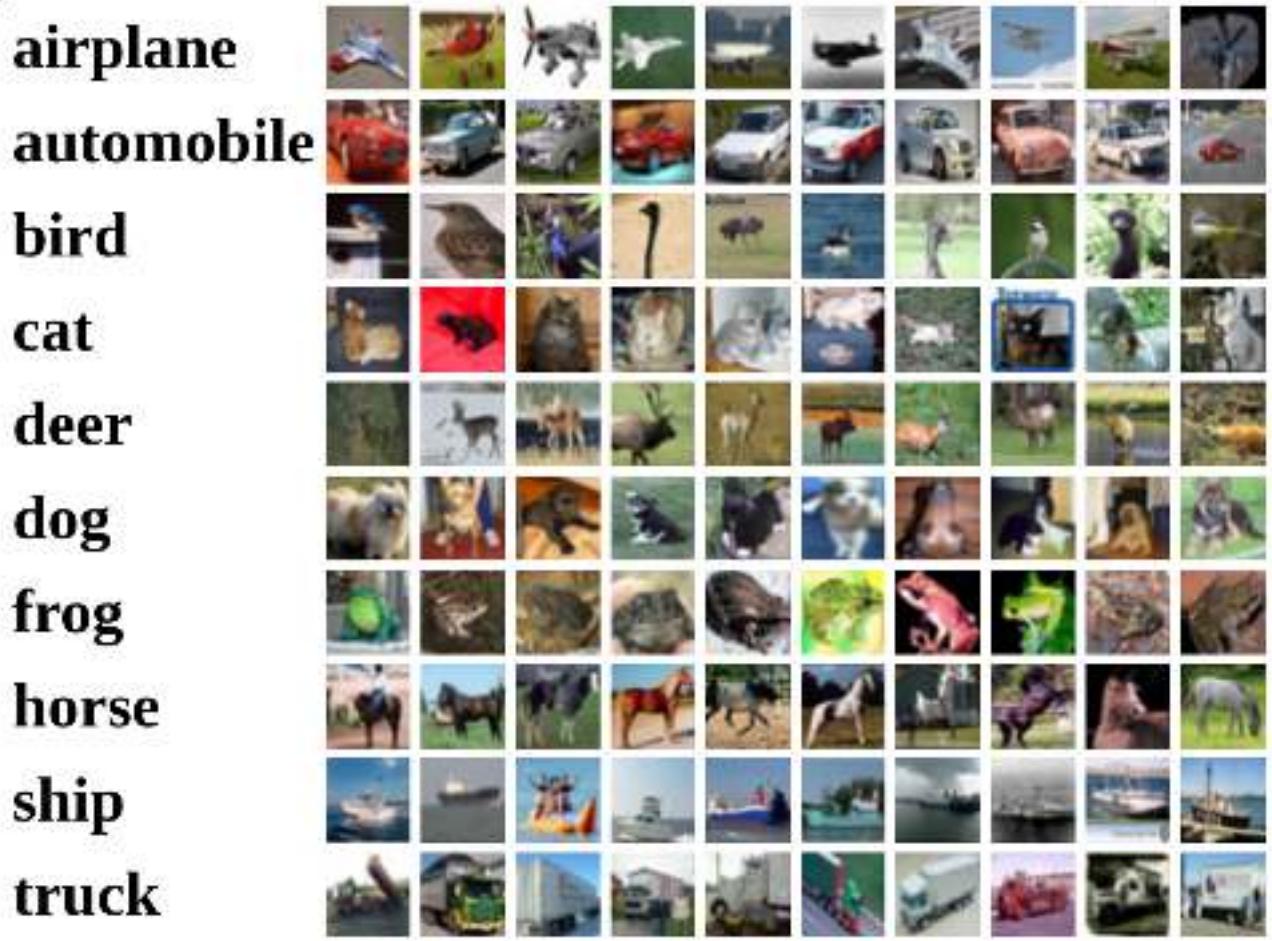


**10 classes**: Digits 0 to 9
**28x28** grayscale images
**50k** training images
**10k** test images

"Drosophila of computer vision"

Results from MNIST often do not hold on more complex datasets!

# Image Classification Datasets: CIFAR10




**10** classes
**50k** training images (5k per class)
**10k** testing images (1k per class)
**32x32 RGB** images

We will use this dataset for homework assignments

Alex Krizhevsky, "Learning Multiple Layers of Features from Tiny Images", Technical Report, 2009.

# Image Classification Datasets: CIFAR100

**100** classes
**50k** training images (500 per class)
**10k** testing images (100 per class)
**32x32 RGB** images

**20 superclasses** with 5 classes each:

Aquatic mammals: beaver, dolphin, otter, seal, whale
Trees: Maple, oak, palm, pine, willow

Alex Krizhevsky, "Learning Multiple Layers of Features from Tiny Images", Technical Report, 2009.

# Image Classification Datasets: ImageNet


**1000** classes

**~1.3M** training images (~1.3K per class)
**50K** validation images (50 per class)
**100K** test images (100 per class)

Performance metric: **Top 5 accuracy**
Algorithm predicts 5 labels for each image; one of them needs to be right

Deng et al, "ImageNet: A Large-Scale Hierarchical Image Database", CVPR 2009
Russakovsky et al, "ImageNet Large Scale Visual Recognition Challenge", IJCV 2015

# Image Classification Datasets: ImageNet


**1000** classes

**~1.3M** training images (~1.3K per class)
**50K** validation images (50 per class)
**100K** test images (100 per class)
test labels are secret!

Images have variable size, but often resized to **256x256** for training

There is also a 22k category version of ImageNet, but less commonly used

Deng et al, "ImageNet: A Large-Scale Hierarchical Image Database", CVPR 2009
Russakovsky et al, "ImageNet Large Scale Visual Recognition Challenge", IJCV 2015

# Image Classification Datasets: MIT Places

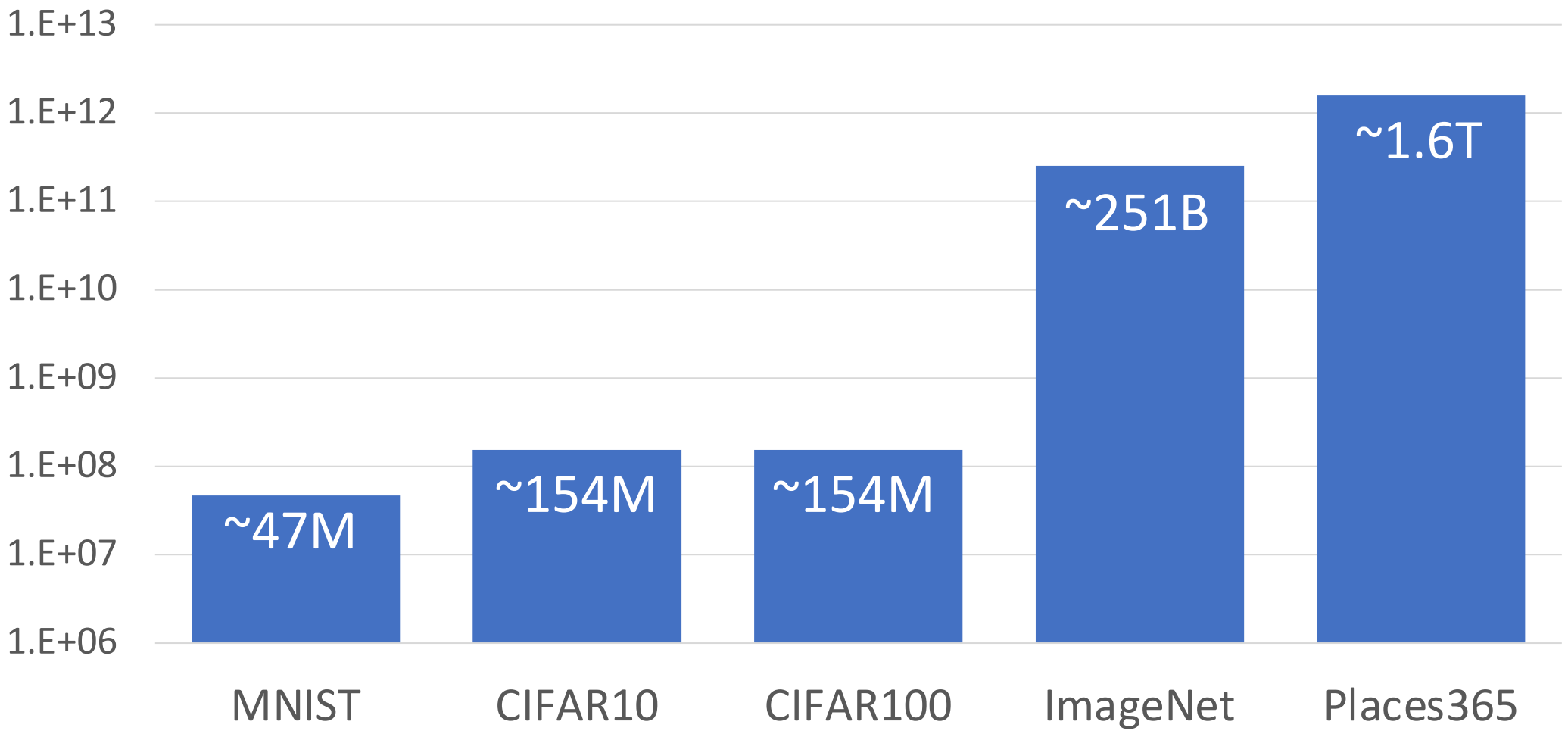

**365 classes** of different scene types

**~8M** training images
**18.25K** val images (50 per class)
**328.5K** test images (900 per class)

Images have variable size, often resize to **256x256** for training

Zhou et al, "Places: A 10 million Image Database for Scene Recognition", TPAMI 2017

# Classification Datasets: Number of Training Pixels



# Image Classification Datasets: **Omniglot**

**1623 categories**: characters from 50 different alphabets

**20 images per category**

Meant to test **few shot learning**

Lake et al, “Human-level concept learning through probabilistic program induction”, Science, 2015

# First classifier: **Nearest Neighbor**

```
def train(images, labels):
  # Machine learning!
  return model
```



Memorize all data and labels

```
def predict(model, test_images):
  # Use model to predict labels
  return test_labels
```

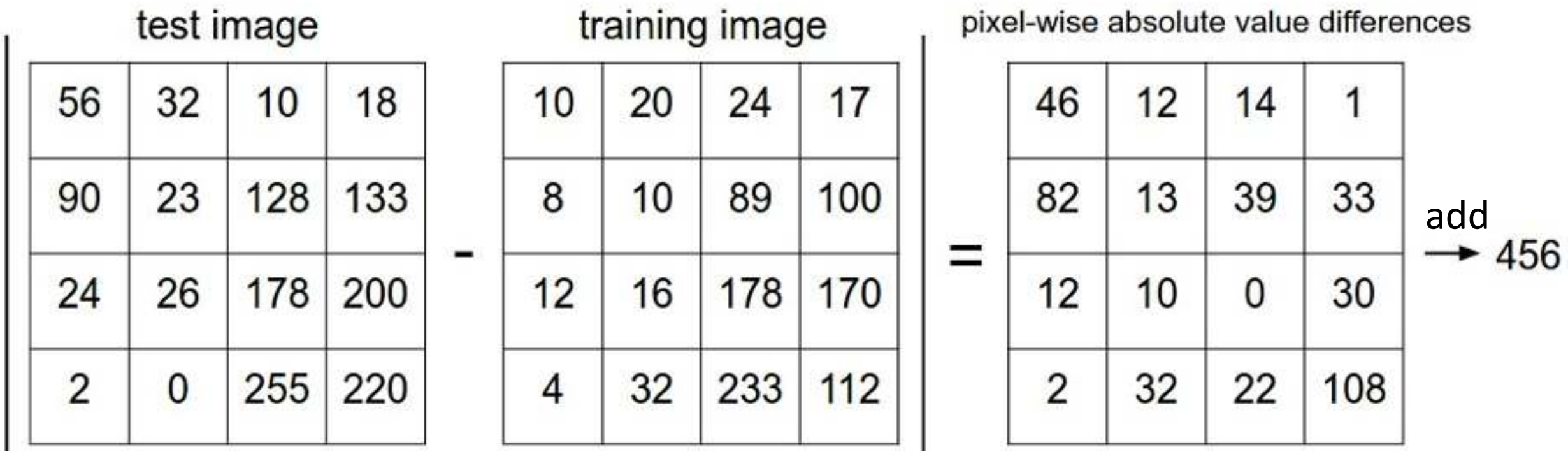


Predict the label of the most similar training image

# Distance Metric to compare images

**L1 distance:**

$$d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$$



# Nearest Neighbor Classifier

```python
import numpy as np

class NearestNeighbor:
  def __init__(self):
    pass

  def train(self, X, y):
    """ X is N x D where each row is an example. Y is 1-dimension of size N """
    # the nearest neighbor classifier simply remembers all the training data
    self.Xtr = X
    self.ytr = y

  def predict(self, X):
    """ X is N x D where each row is an example we wish to predict label for """
    num_test = X.shape[0]
    # lets make sure that the output type matches the input type
    Ypred = np.zeros(num_test, dtype = self.ytr.dtype)

    # loop over all test rows
    for i in xrange(num_test):
      # find the nearest training image to the i'th test image
      # using the L1 distance (sum of absolute value differences)
      distances = np.sum(np.abs(self.Xtr - X[i,:]), axis = 1)
      min_index = np.argmin(distances) # get the index with smallest distance
      Ypred[i] = self.ytr[min_index] # predict the label of the nearest example

    return Ypred
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```

Memorize training data

# Nearest Neighbor Classifier

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      Ypred[i] = self.ytr[min_index] # predict the label of the nearest example

    return Ypred
```

For each test image:

Find nearest training image

Return label of nearest image

```python
import numpy as np

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# Nearest Neighbor Classifier

**Q**: With N examples, how fast is training?

```python
import numpy as np

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# Nearest Neighbor Classifier

**Q**: With N examples, how fast is training?

**A**: O(1)

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    return Ypred
```

# Nearest Neighbor Classifier

**Q**: With N examples, how fast is training?

**A**: O(1)

**Q**: With N examples, how fast is testing?

```python
import numpy as np

class NearestNeighbor:
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    pass

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      min_index = np.argmin(distances) # get the index with smallest distance
      Ypred[i] = self.ytr[min_index] # predict the label of the nearest example

    return Ypred
```

# Nearest Neighbor Classifier

**Q**: With N examples, how fast is training?

**A**: O(1)

**Q**: With N examples, how fast is testing?

**A**: O(N)

```python
import numpy as np

class NearestNeighbor:
  def __init__(self):
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      min_index = np.argmin(distances) # get the index with smallest distance
      Ypred[i] = self.ytr[min_index] # predict the label of the nearest example

    return Ypred
```

# Nearest Neighbor Classifier

**Q**: With N examples, how fast is training?

**A**: O(1)

**Q**: With N examples, how fast is testing?

**A**: O(N)

This is **bad**: We can afford slow training, but we need fast testing!

# Nearest Neighbor Classifier

```python
import numpy as np

class NearestNeighbor:
  def __init__(self):
    pass

  def train(self, X, y):
    """ X is N x D where each row is an example. Y is 1-dimension of size N """
    # the nearest neighbor classifier simply remembers all the training data
    self.Xtr = X
    self.ytr = y

  def predict(self, X):
    """ X is N x D where each row is an example we wish to predict label for """
    num_test = X.shape[0]
    # lets make sure that the output type matches the input type
    Ypred = np.zeros(num_test, dtype = self.ytr.dtype)

    # loop over all test rows
    for i in xrange(num_test):
      # find the nearest training image to the i'th test image
      # using the L1 distance (sum of absolute value differences)
      distances = np.sum(np.abs(self.Xtr - X[i,:]), axis = 1)
      min_index = np.argmin(distances) # get the index with smallest distance
      Ypred[i] = self.ytr[min_index] # predict the label of the nearest example

    return Ypred
```

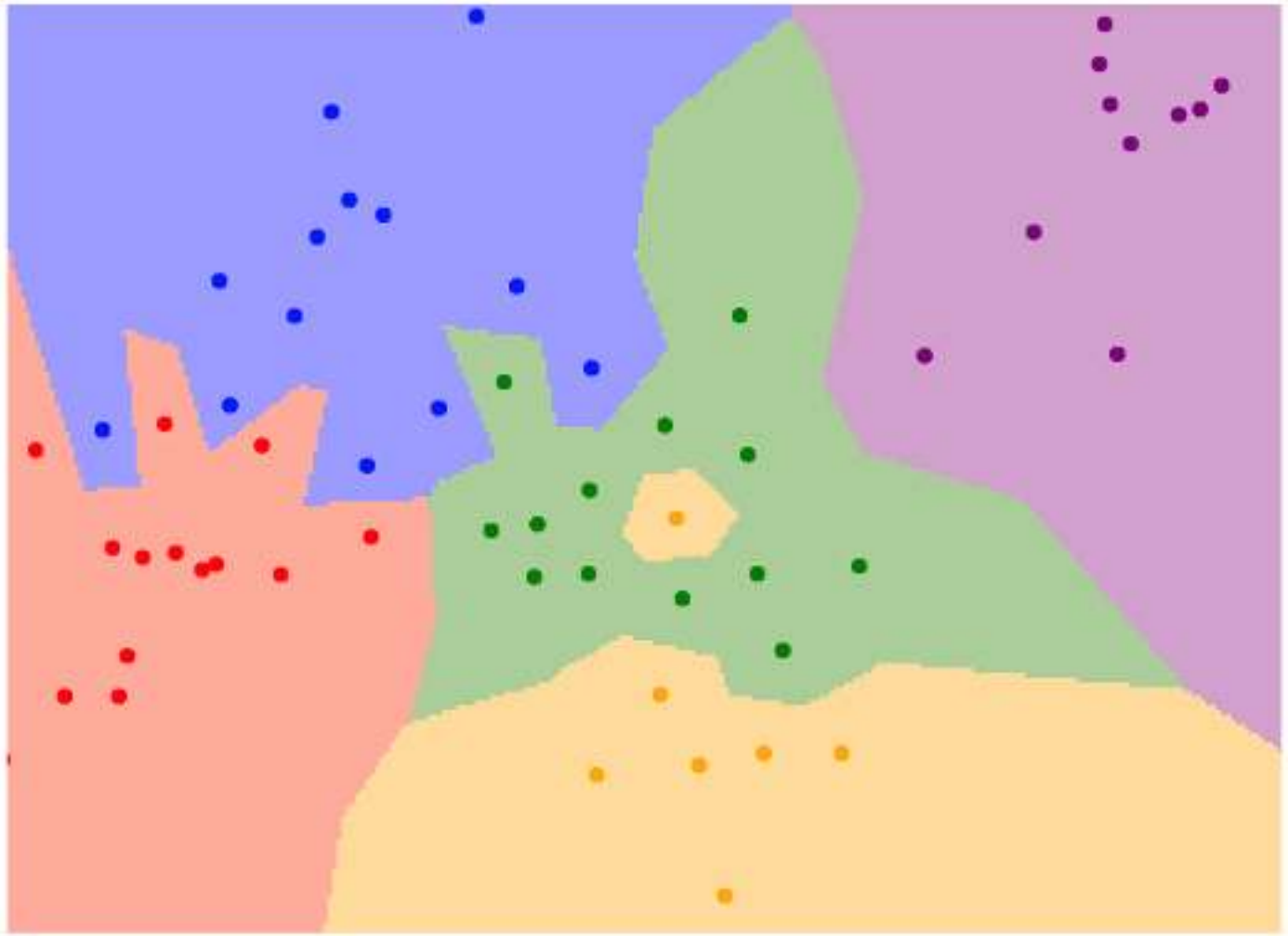
There are many methods for fast / approximate nearest neighbors; e.g. see

https://github.com/facebookresearch/faiss

# What does this look like?

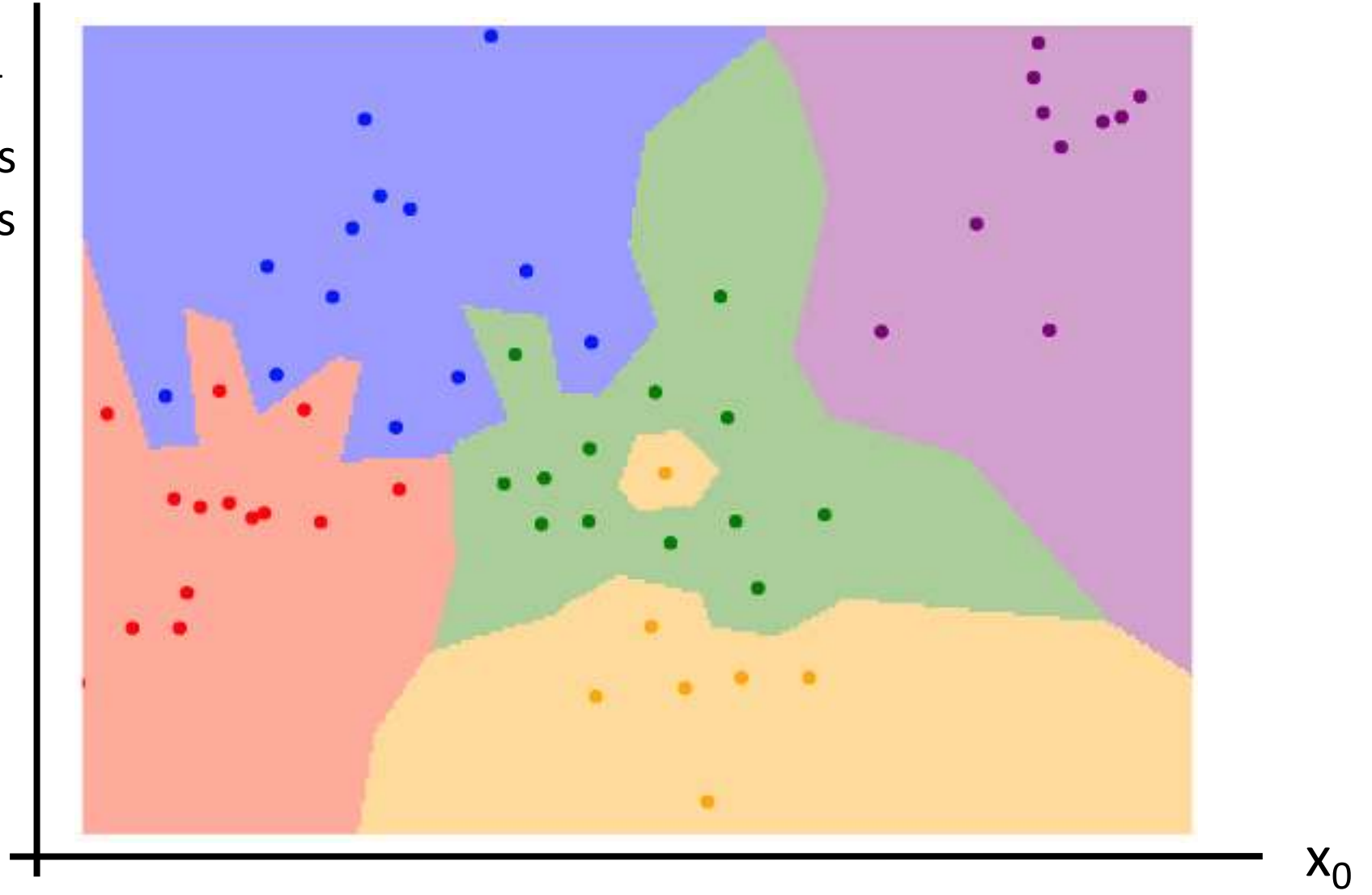
# What does this look like?

# Nearest Neighbor **Decision Boundaries**



# Nearest Neighbor **Decision Boundaries**

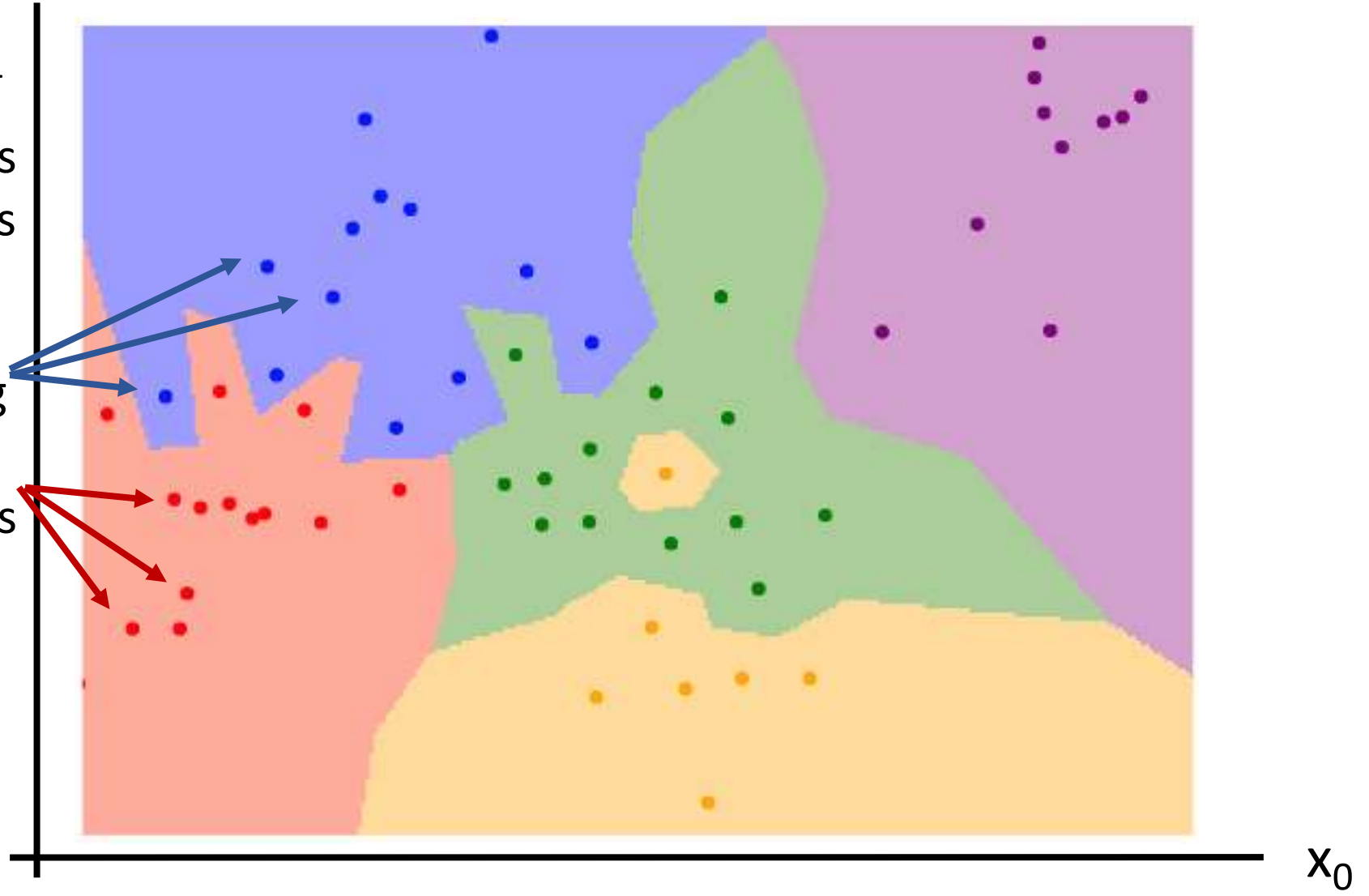
Nearest neighbors in two dimensions



# Nearest Neighbor **Decision Boundaries**

Nearest neighbors in two dimensions

Points are training examples; colors give training labels

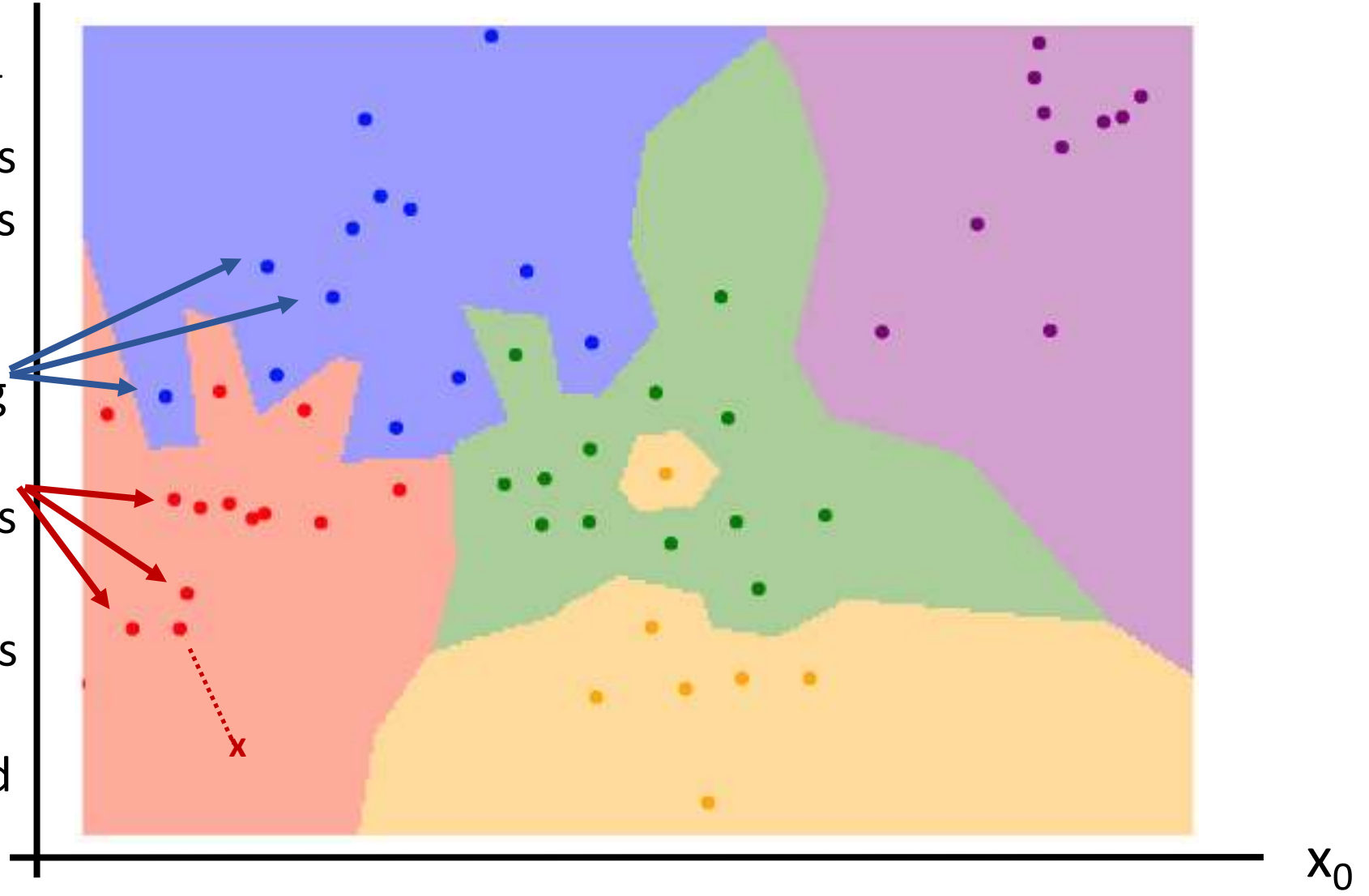


# Nearest Neighbor **Decision Boundaries**

Nearest neighbors in two dimensions

Points are training examples; colors give training labels

Background colors give the category a test point would be assigned



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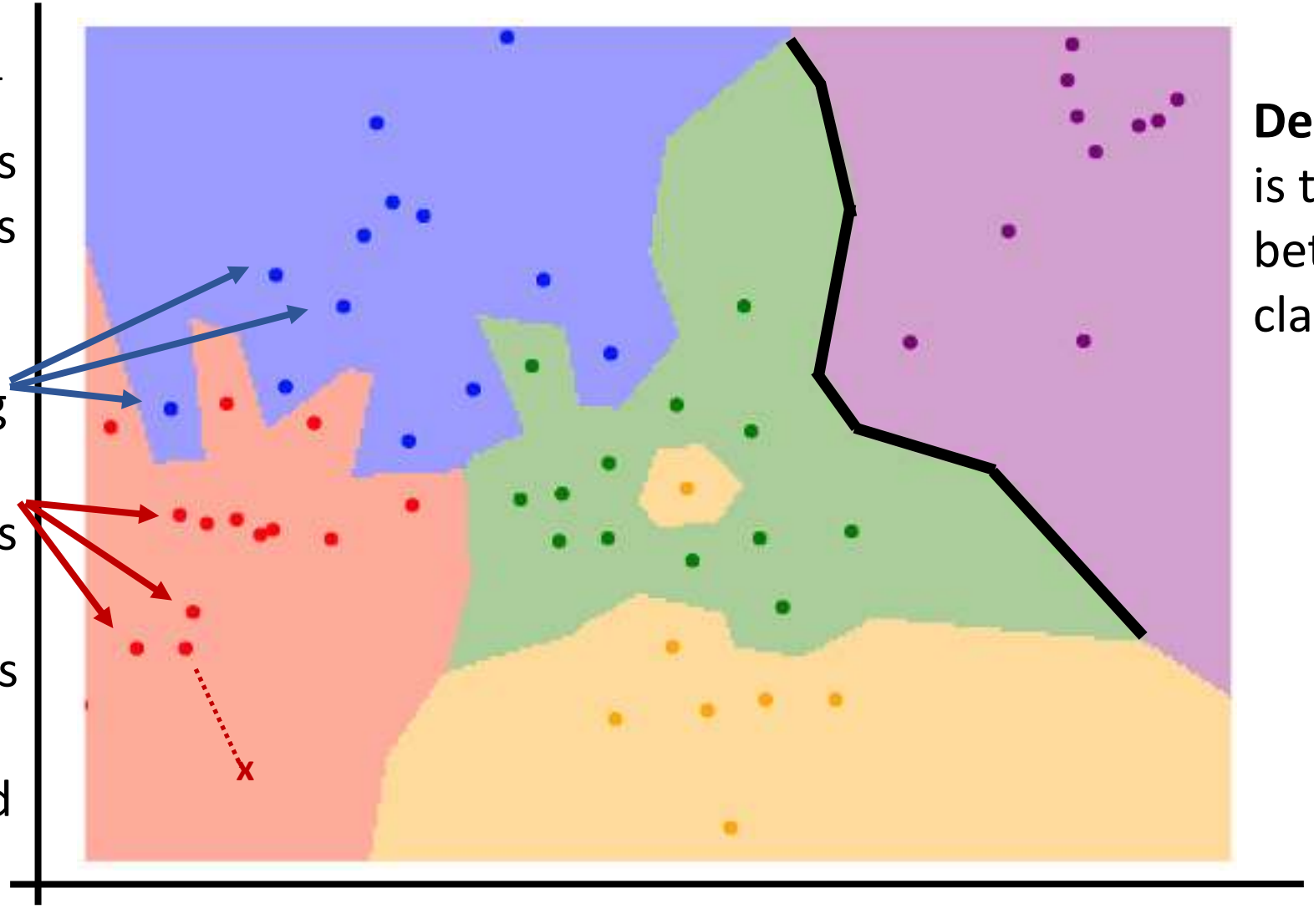


**Decision boundary** is the boundary between two classification regions

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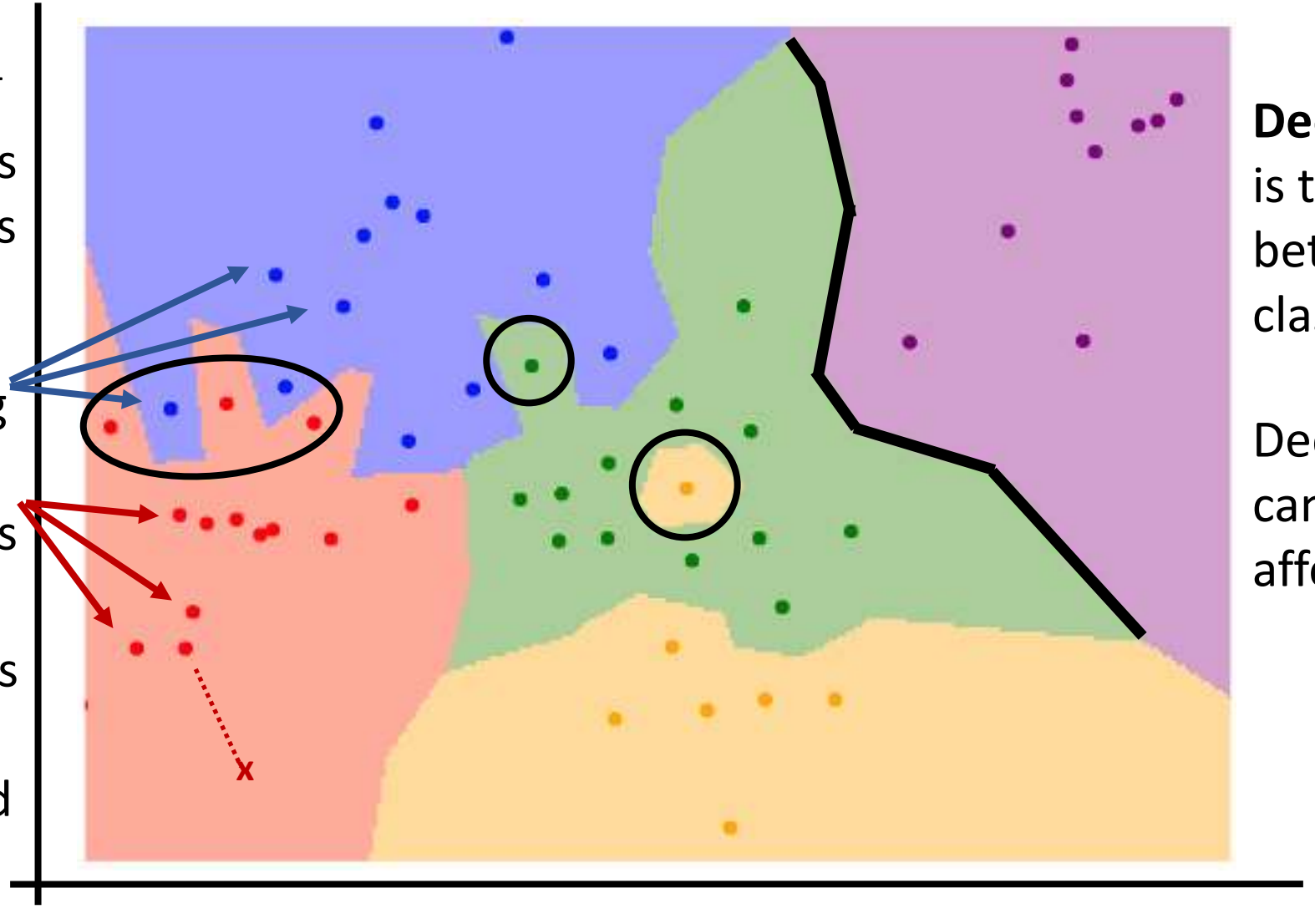
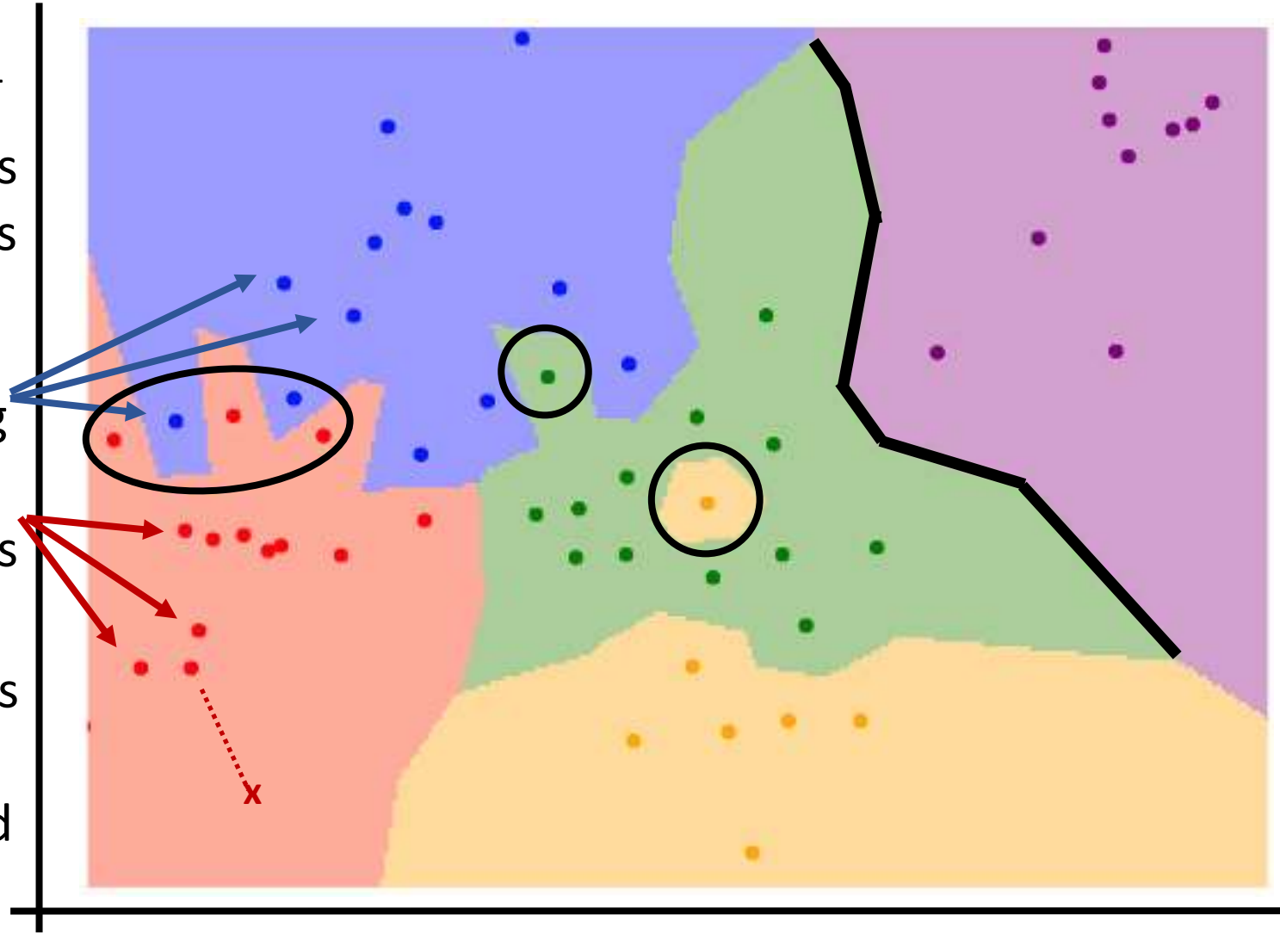



**Decision boundary** is the boundary between two classification regions

Decision boundaries can be noisy; affected by outliers

# Nearest Neighbor **Decision Boundaries**




Nearest neighbors in two dimensions

Points are training examples; colors give training labels

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**Decision boundary** is the boundary between two classification regions

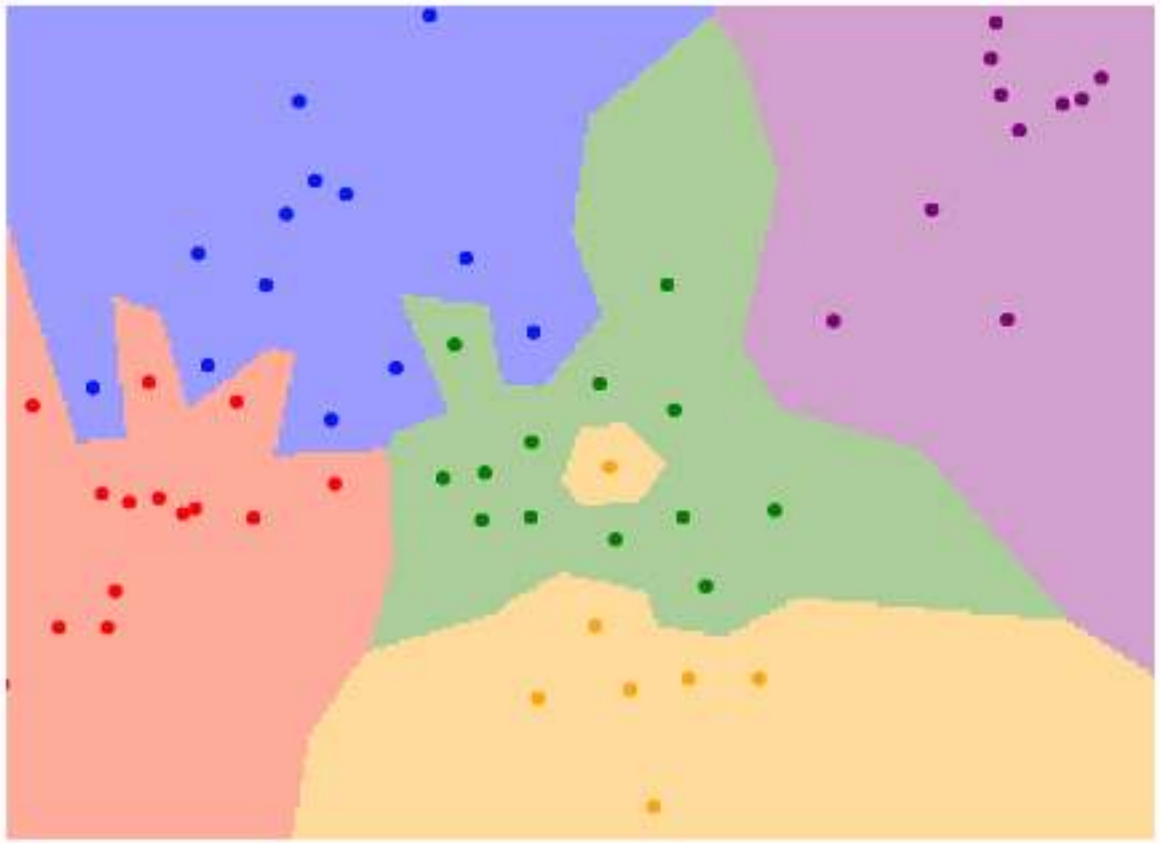
Decision boundaries can be noisy; affected by outliers

How to smooth out decision boundaries? Use more neighbors!

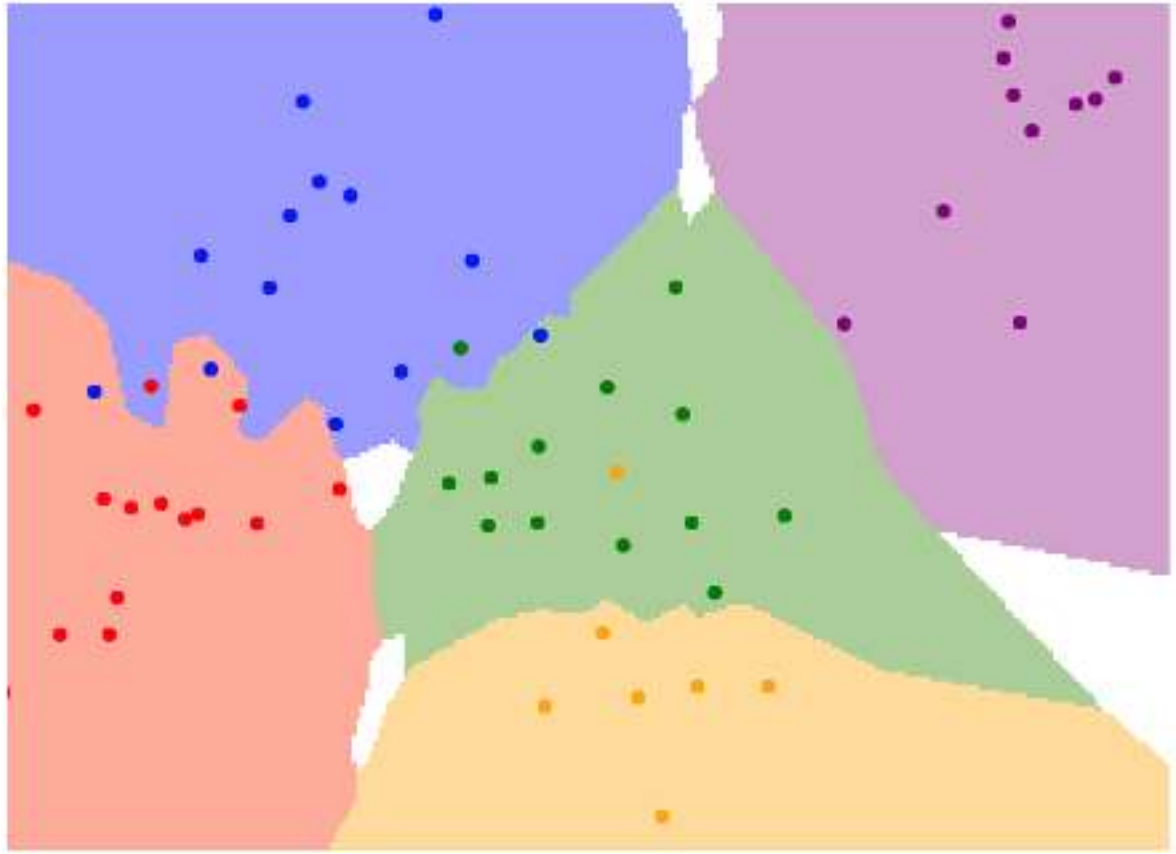
# K-Nearest Neighbors

Instead of copying label from nearest neighbor, take **majority vote** from K closest points

K = 1



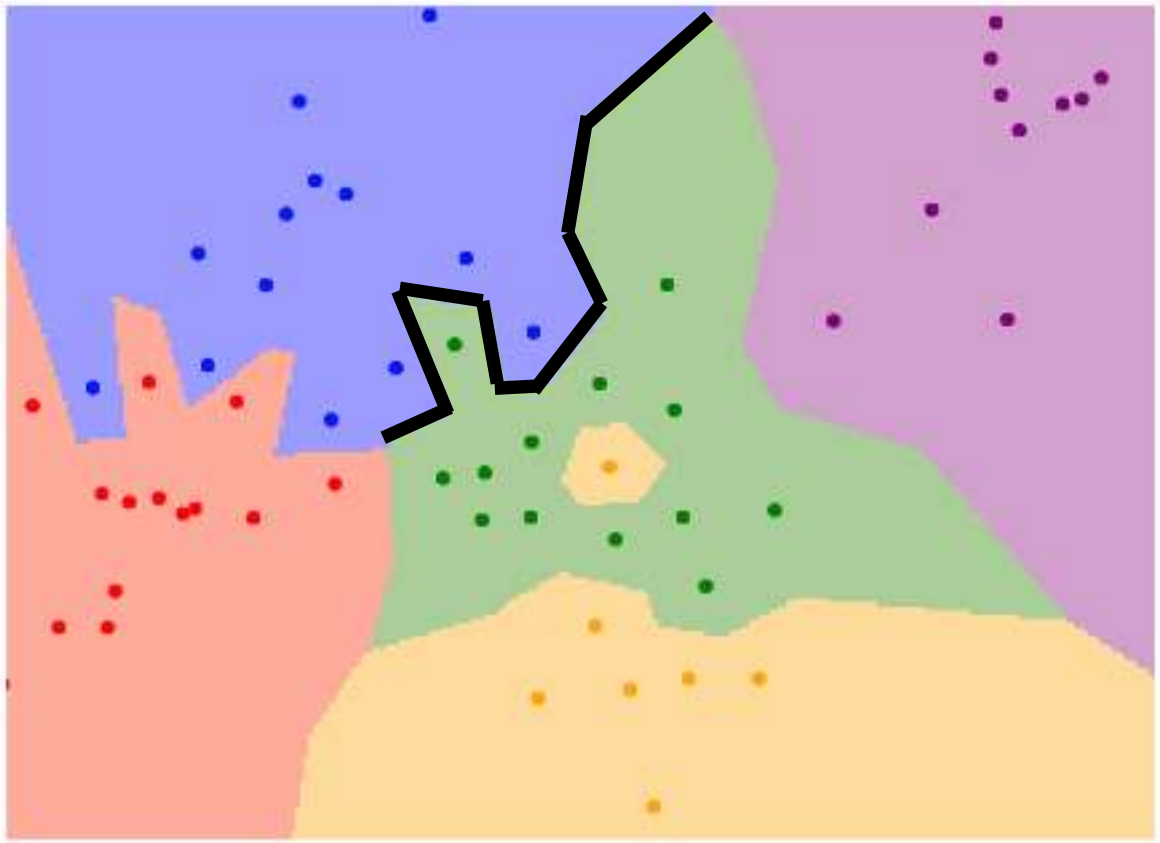
K = 3



# K-Nearest Neighbors

Using more neighbors helps smooth out rough decision boundaries

K = 1



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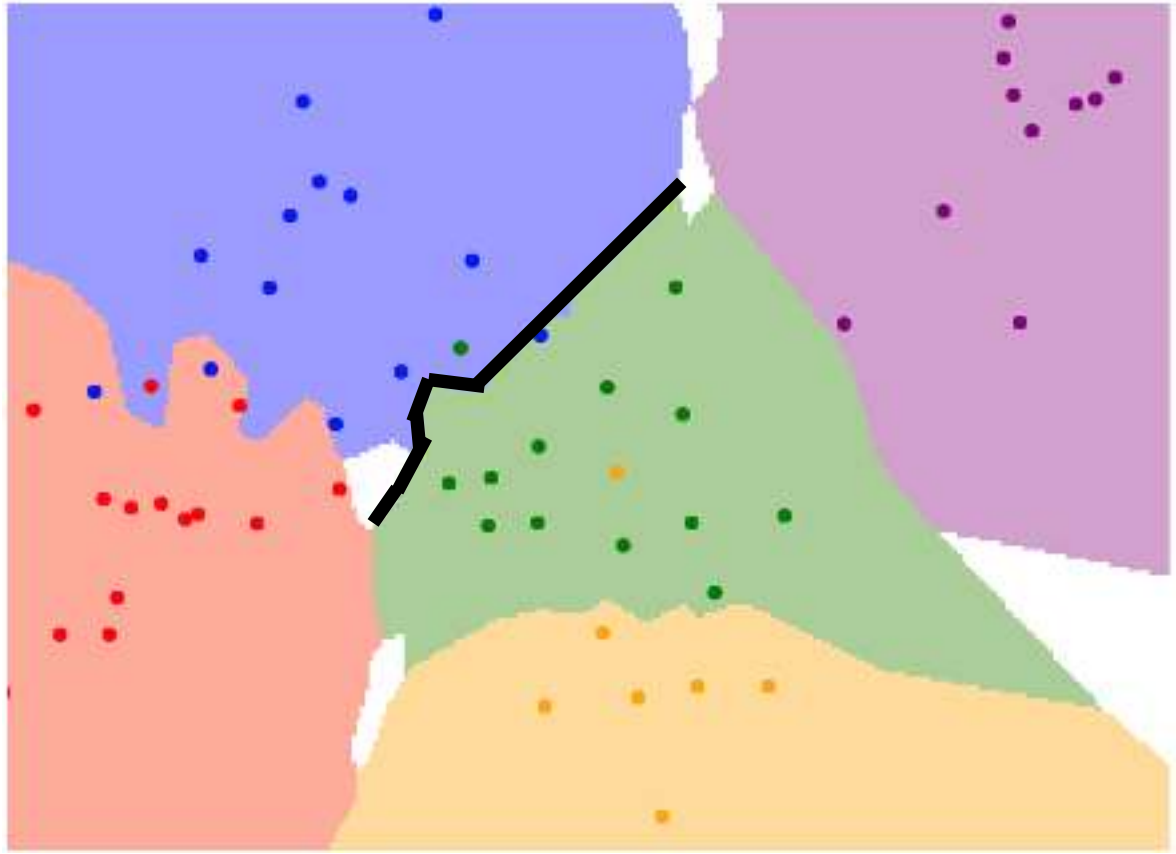
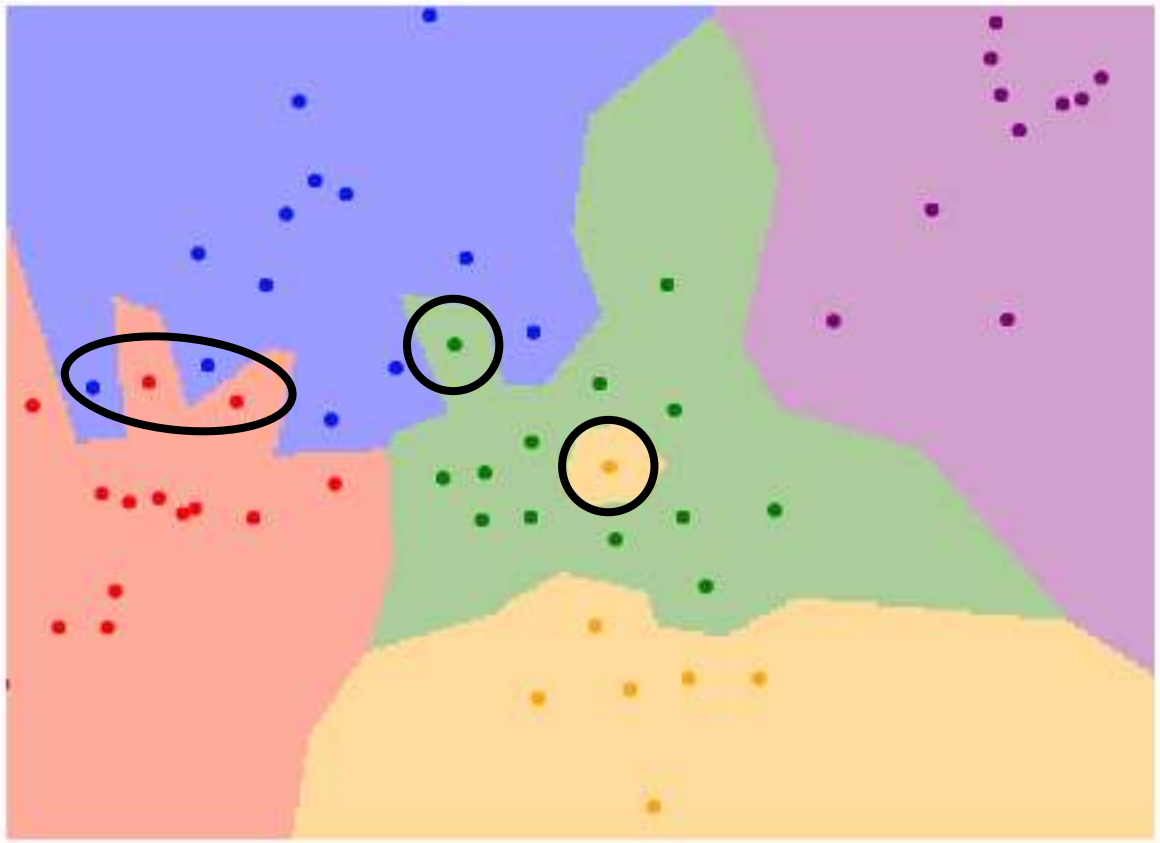


# K-Nearest Neighbors

Using more neighbors helps reduce the effect of outliers

K = 1



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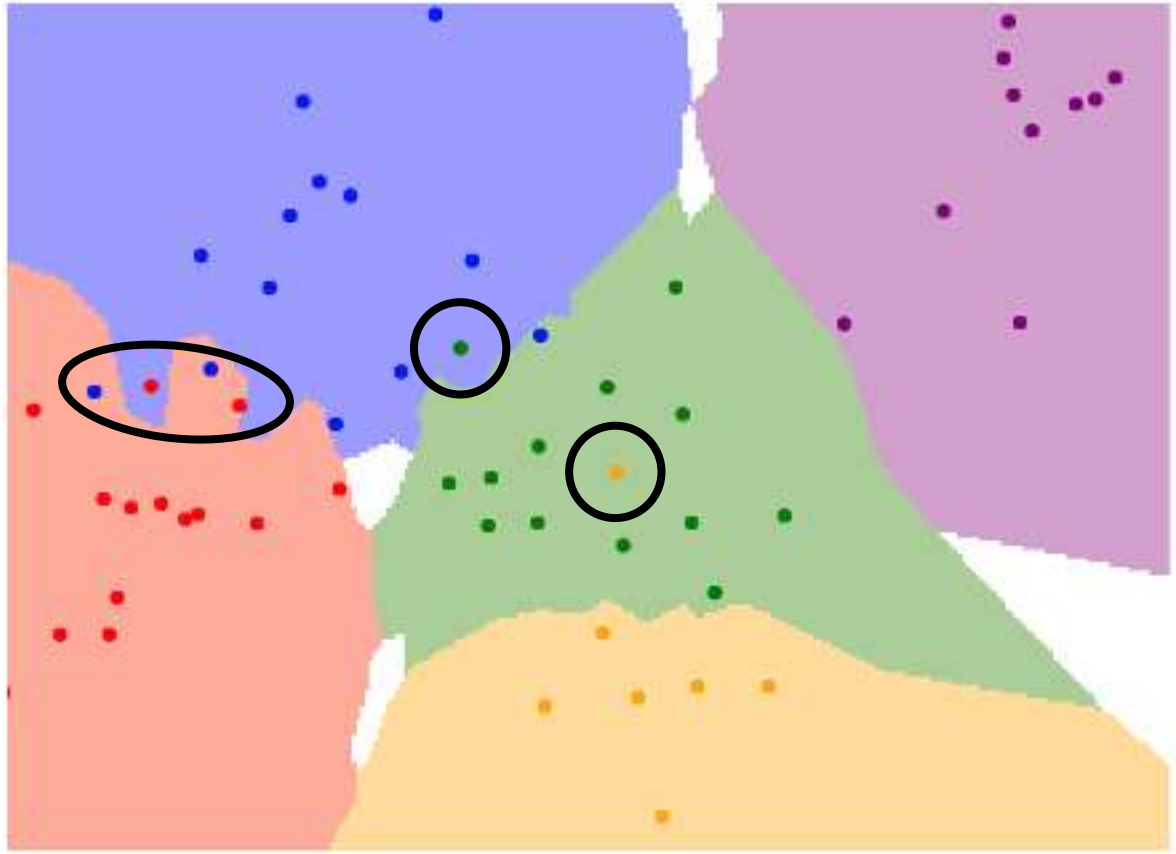
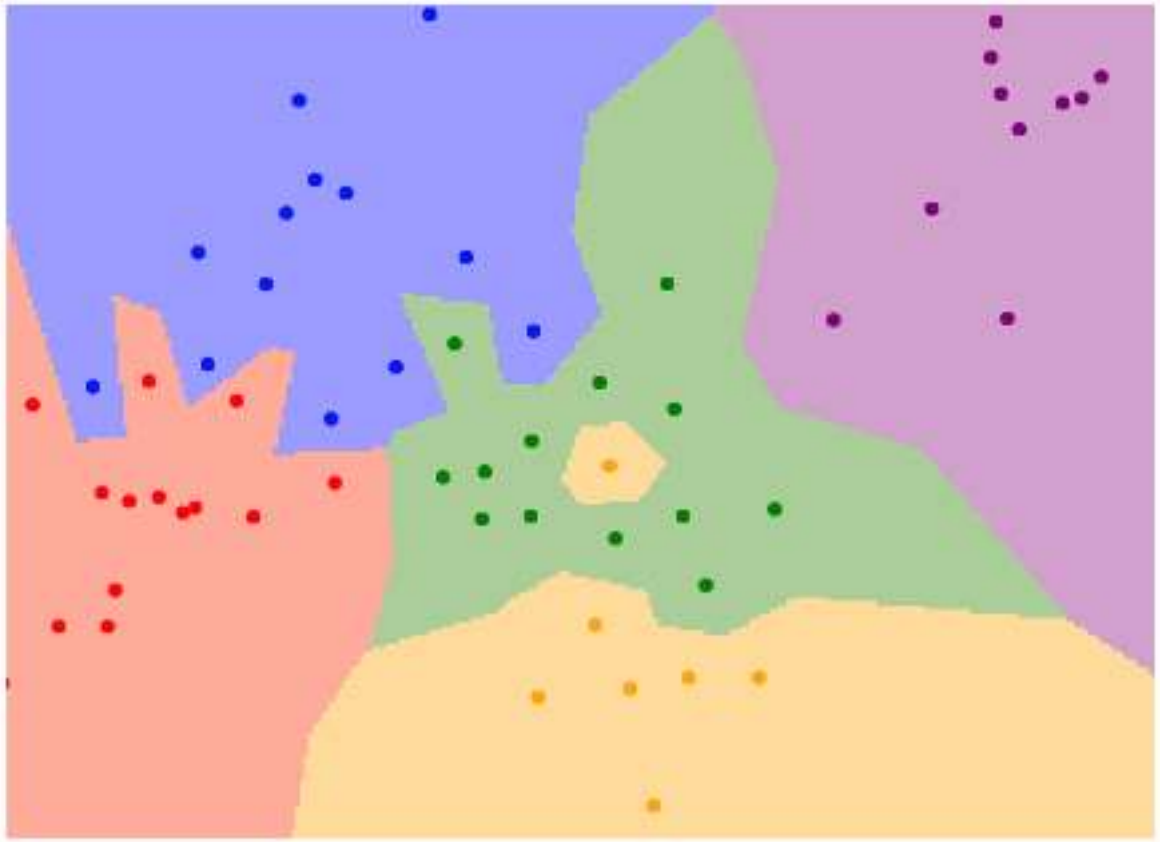


# K-Nearest Neighbors

When K > 1 there can be ties between classes.
Need to break somehow!

K = 1



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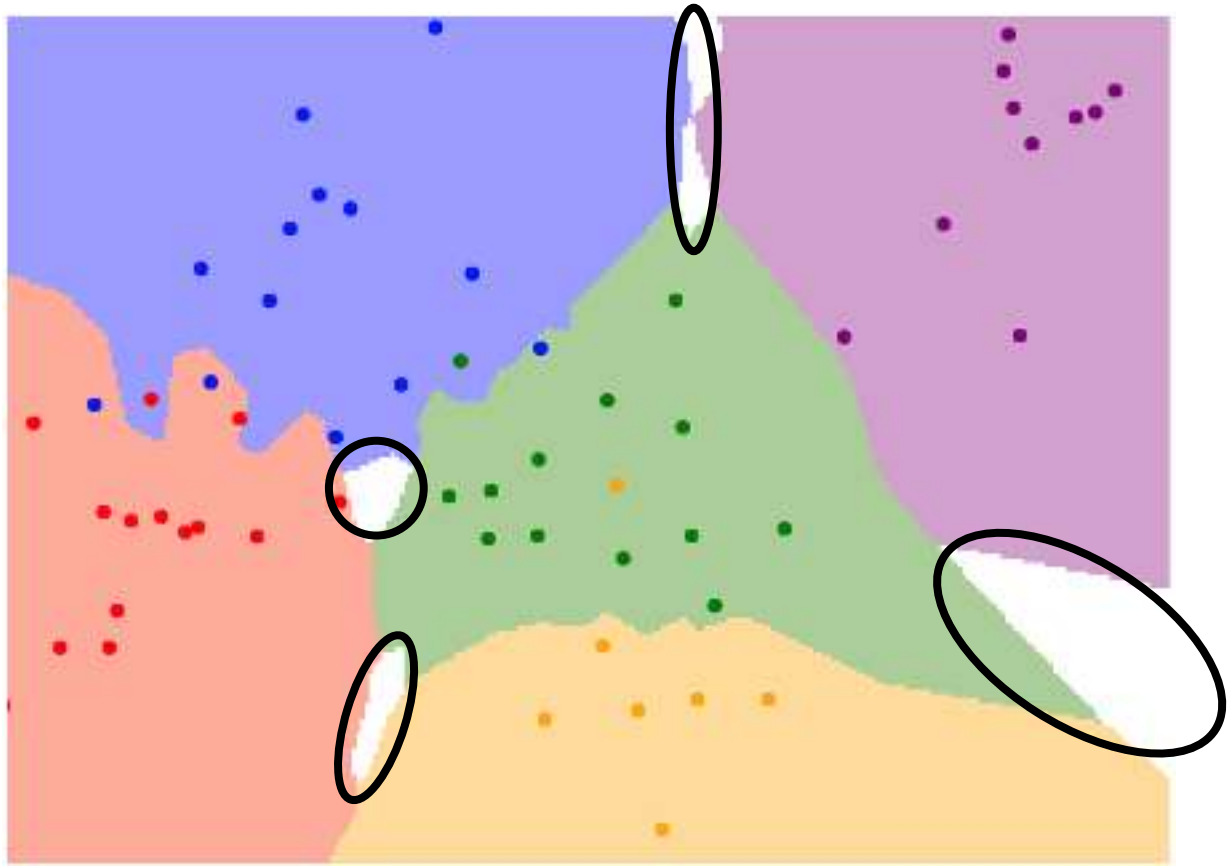
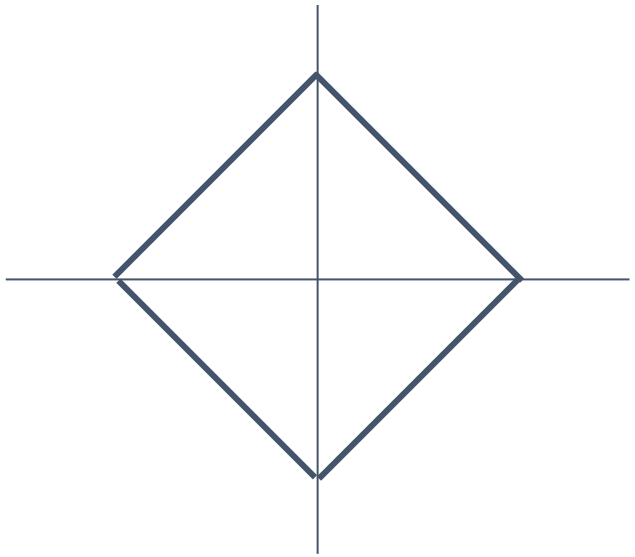


# K-Nearest Neighbors: Distance Metric

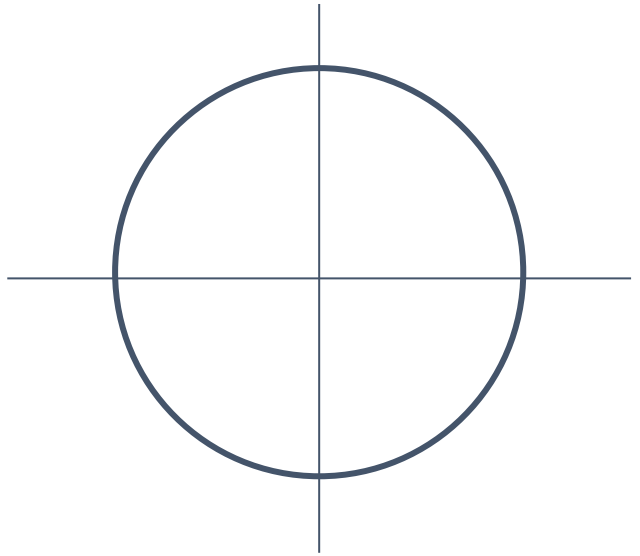
L1 (Manhattan) distance

$$d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$$



L2 (Euclidean) distance

$$d_1(I_1, I_2) = \left( \sum_p (I_1^p - I_2^p)^2 \right)^{\frac{1}{2}}$$



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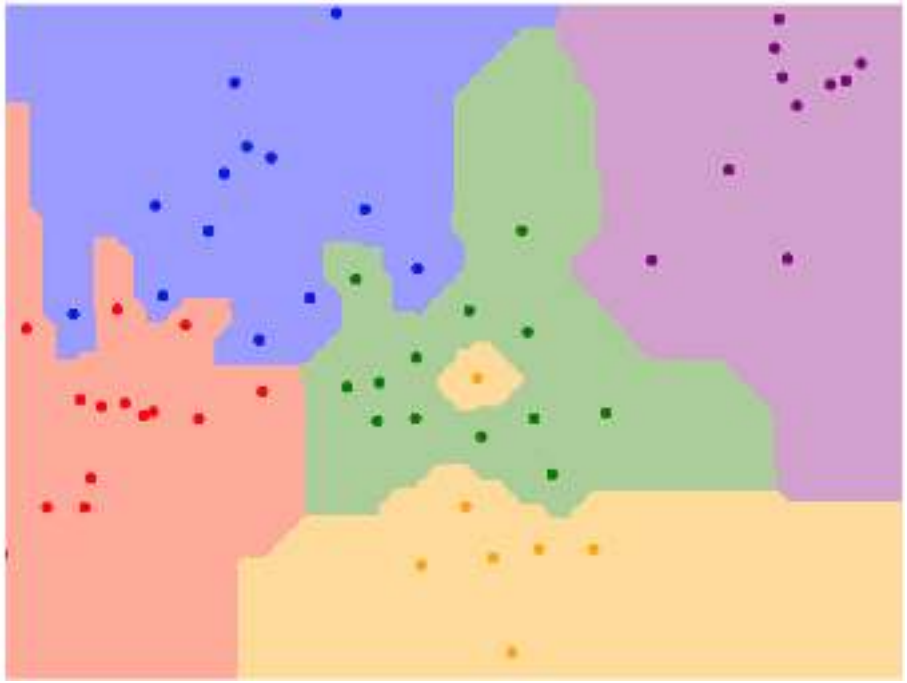

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# K-Nearest Neighbors: Distance Metric

With the right choice of distance metric, we can apply K-Nearest Neighbor to any type of data!

# K-Nearest Neighbors: **Distance Metric**

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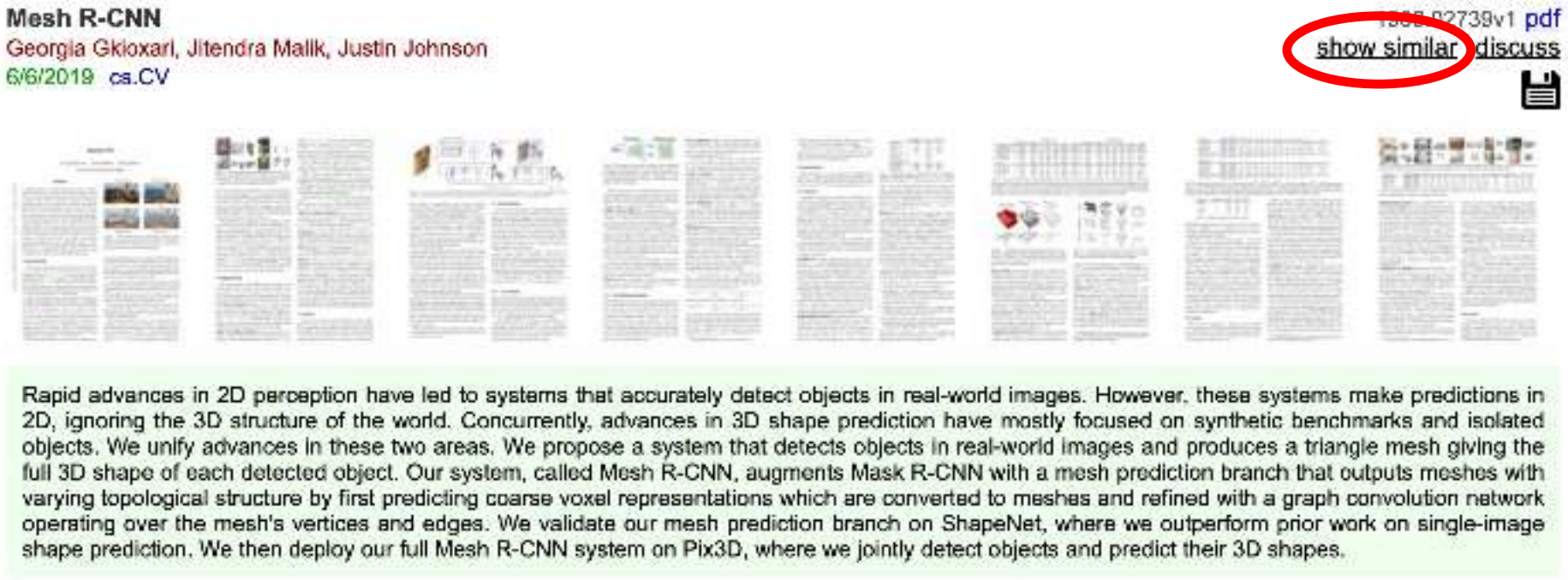
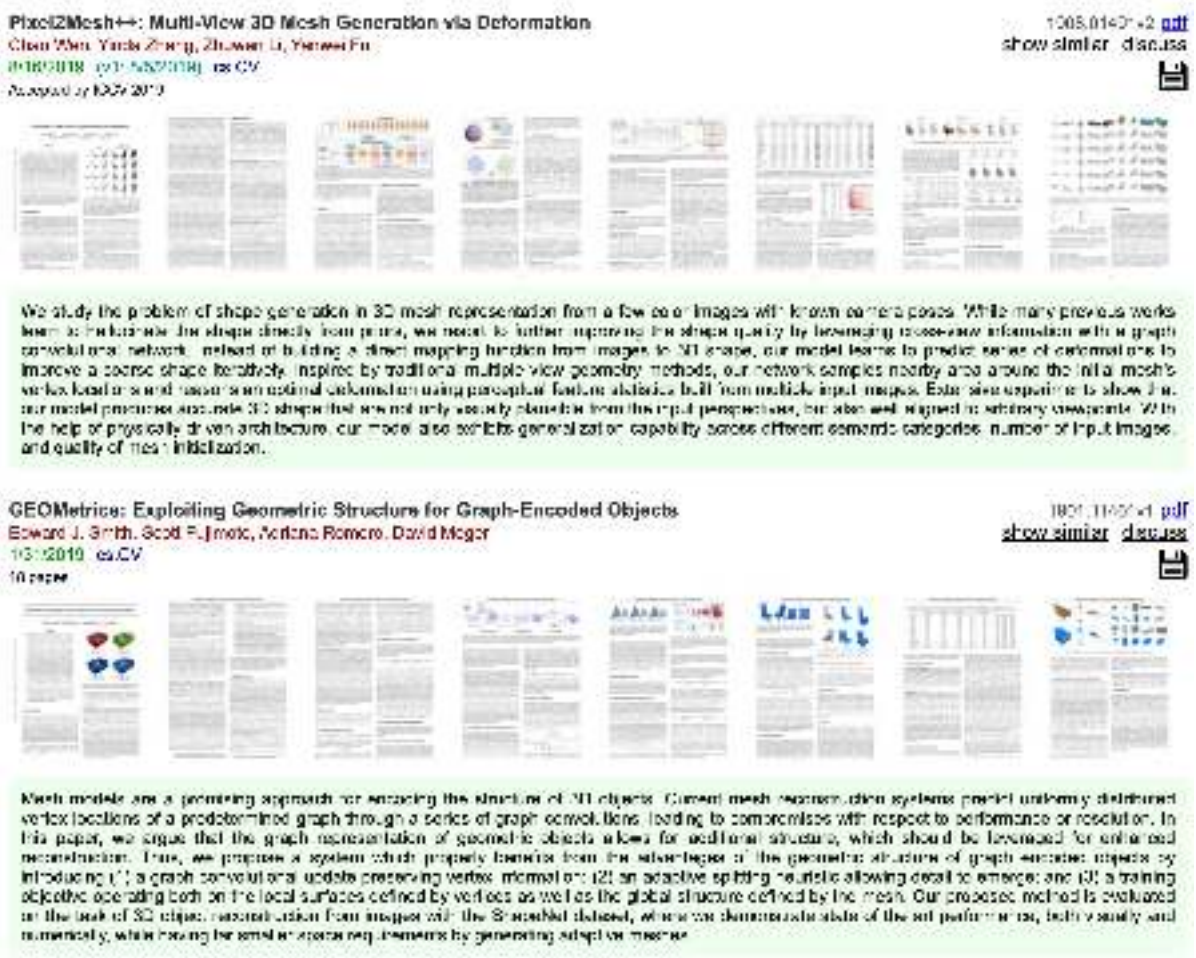
Example: Compare research papers using tf-idf similarity




http://www.arxiv-sanity.com/search?q=mesh+r-cnn

# K-Nearest Neighbors: **Distance Metric**

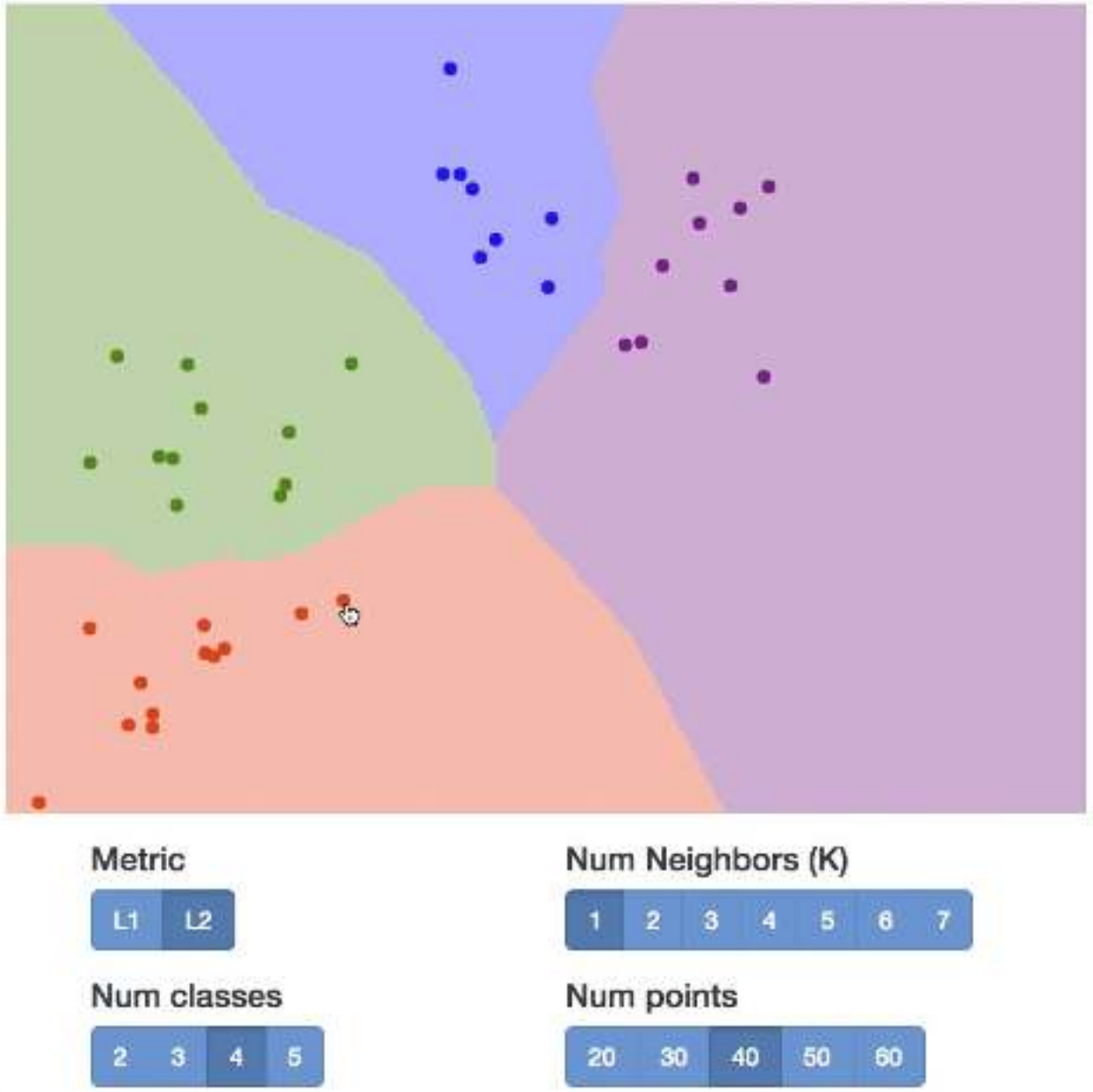

http://www.arxiv-sanity.com/1906.02739v1

# K-Nearest Neighbors: Web Demo

Interactively move points around and see decision boundaries change

Play with L1 vs L2 metrics

Play with changing number of training points, value of K




http://vision.stanford.edu/teaching/cs231n-demos/knn/

# Hyperparameters

What is the best value of **K** to use?
What is the best **distance metric** to use?

These are examples of **hyperparameters**: choices about our learning algorithm that we don't learn from the training data; instead we set them at the start of the learning process

# Hyperparameters

What is the best value of **K** to use?
What is the best **distance metric** to use?

These are examples of **hyperparameters**: choices about our learning algorithm that we don't learn from the training data; instead we set them at the start of the learning process

Very problem-dependent. In general need to try them all and see what works best for our data / task.

# Setting Hyperparameters

**Idea #1**: Choose hyperparameters that work best on the data

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**BAD**: K = 1 always works perfectly on training data

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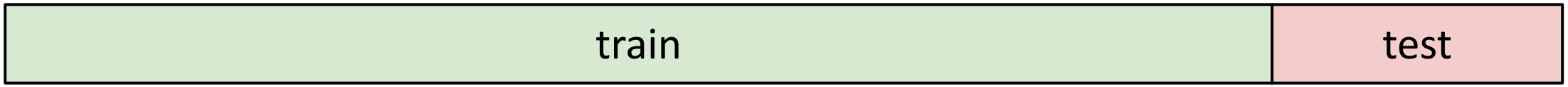
**BAD**: K = 1 always works perfectly on training data


**Idea #2**: Split data into **train** and **test**, choose hyperparameters that work best on test data



# Setting Hyperparameters

**Idea #1**: Choose hyperparameters that work best on the data

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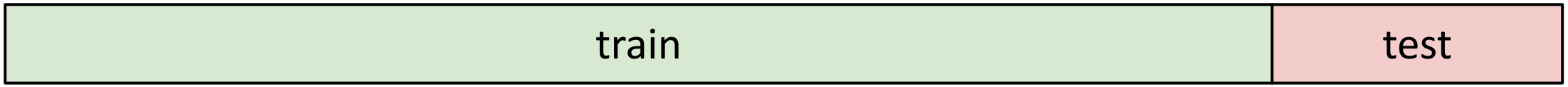
**Idea #2**: Split data into **train** and **test**, choose hyperparameters that work best on test data

**BAD**: No idea how algorithm will perform on new data



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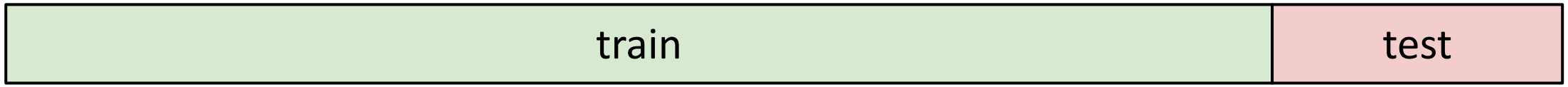
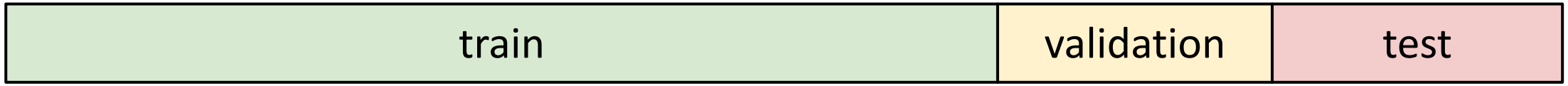


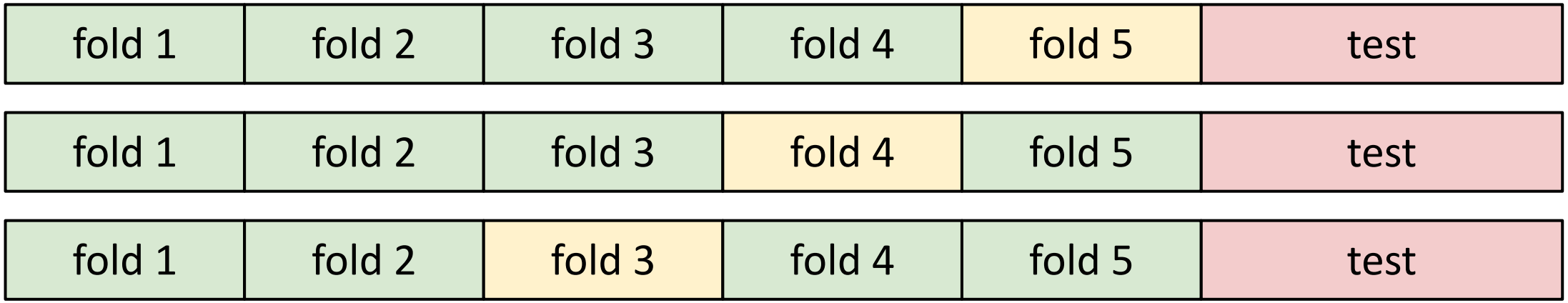

**Idea #3**: Split data into **train**, **val**, and **test**; choose hyperparameters on val and evaluate on test

**Better!**



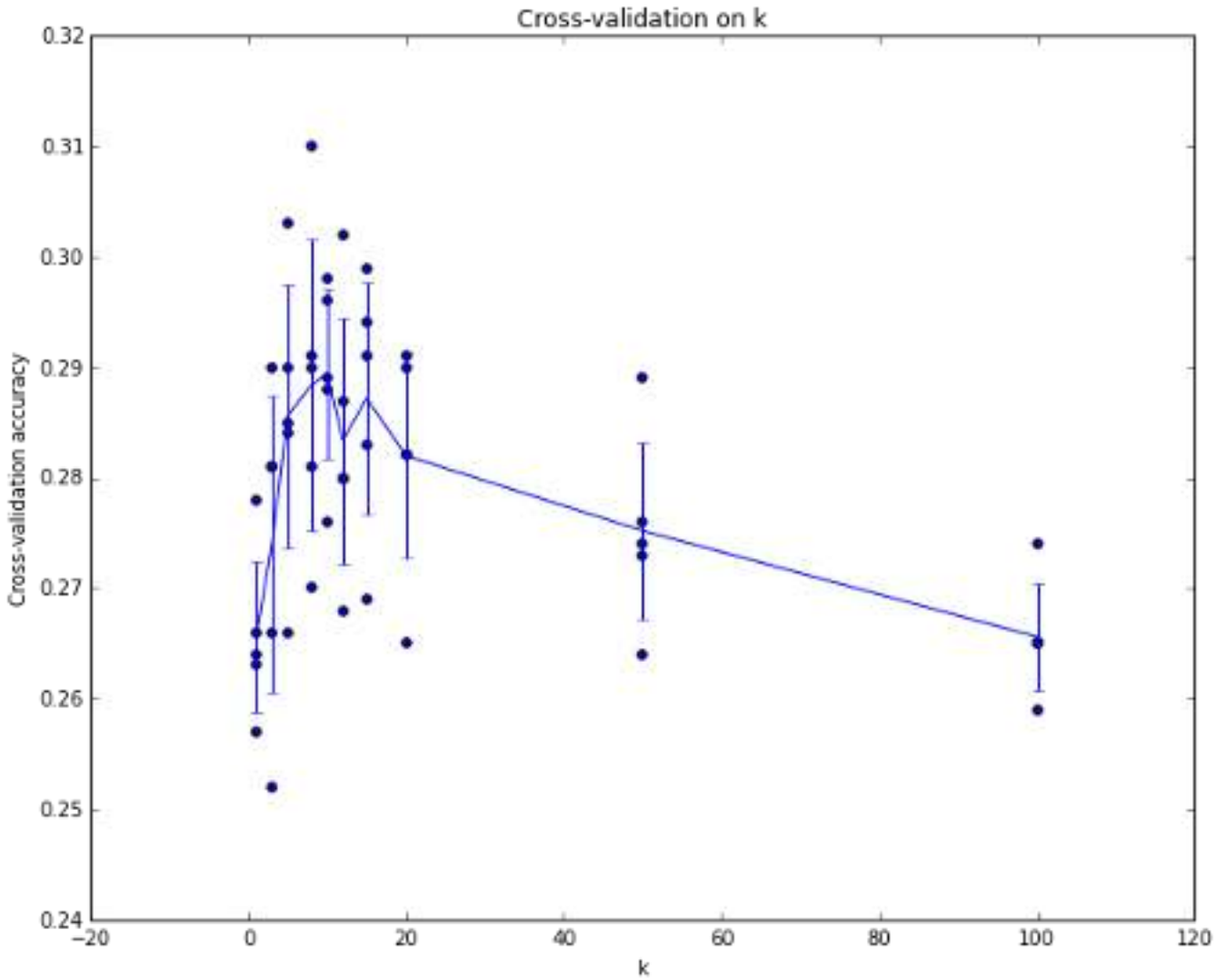
# Setting Hyperparameters


**Idea #4**: **Cross-Validation**: Split data into **folds**, try each fold as validation and average the results




Useful for small datasets, but (unfortunately) not used too frequently in deep learning

# Setting Hyperparameters




Example of 5-fold cross-validation for the value of **k.**

Each point: single outcome.

The line goes through the mean, bars indicated standard deviation

(Seems that k ~ 7 works best for this data)

# K-Nearest Neighbor: **Universal Approximation**

As the number of training samples goes to infinity, nearest neighbor can represent any(*) function!

(*) Subject to many technical conditions. Only continuous functions on a compact domain; need to make assumptions about spacing of training points; etc.

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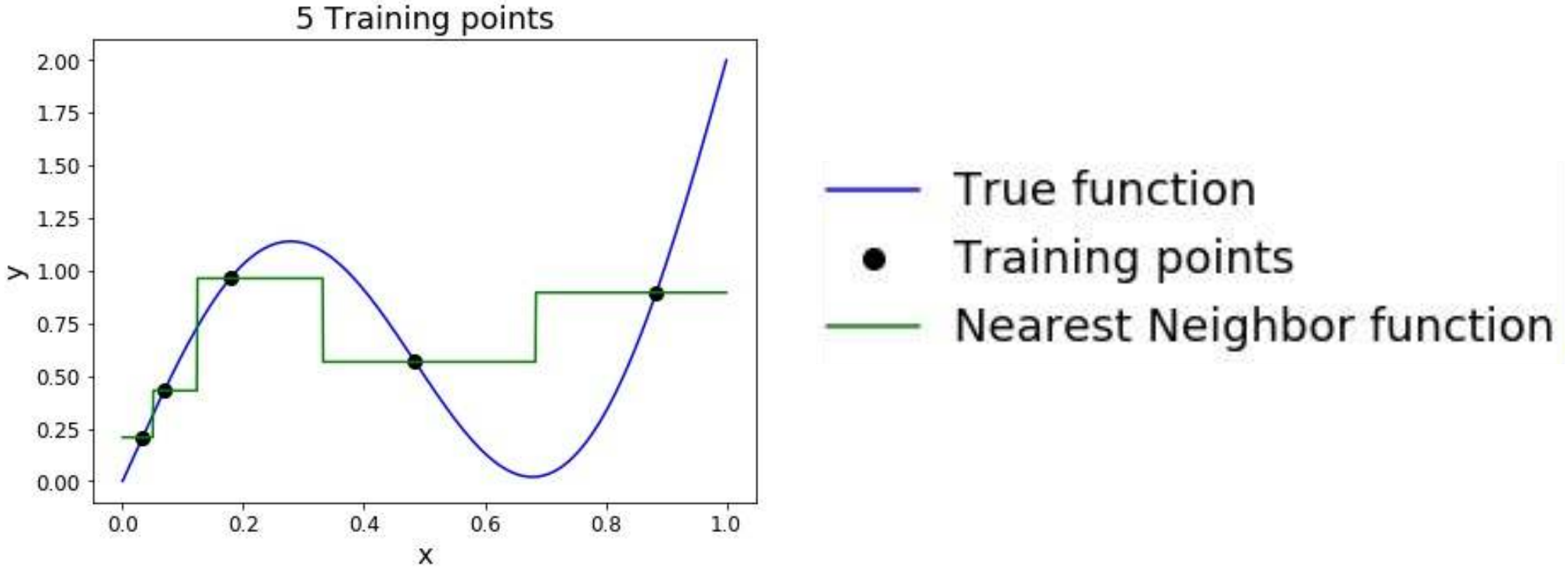


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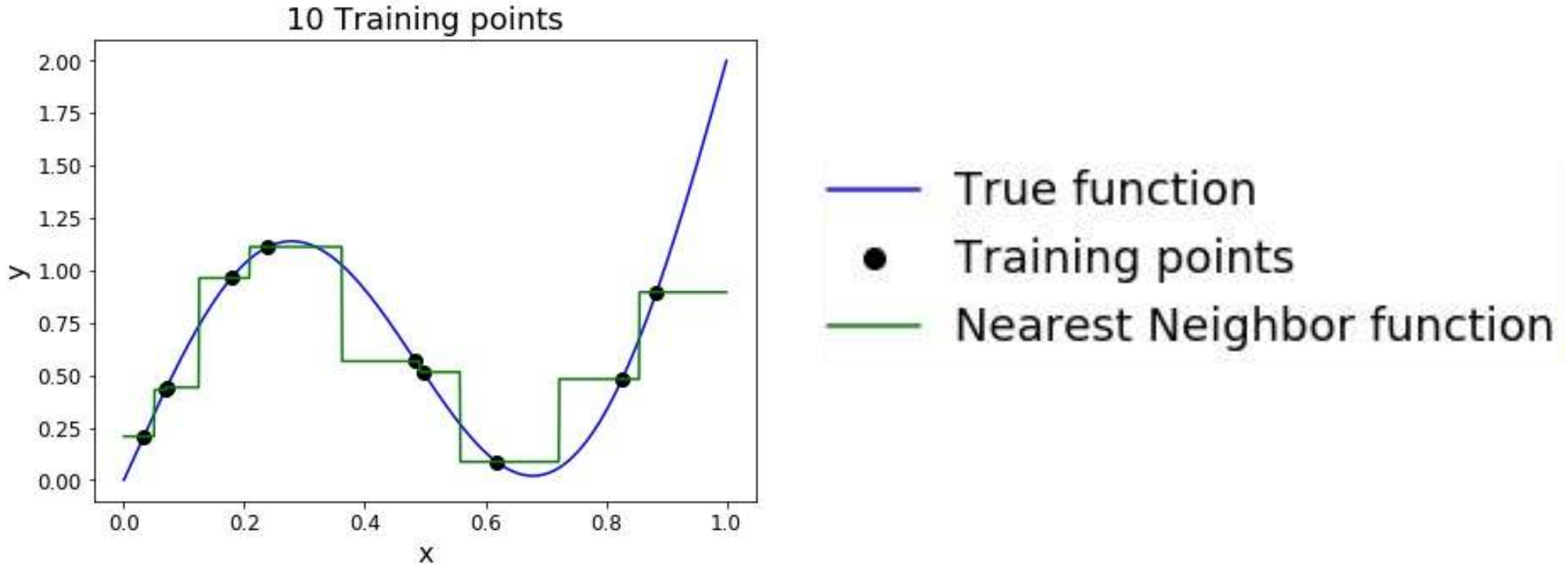


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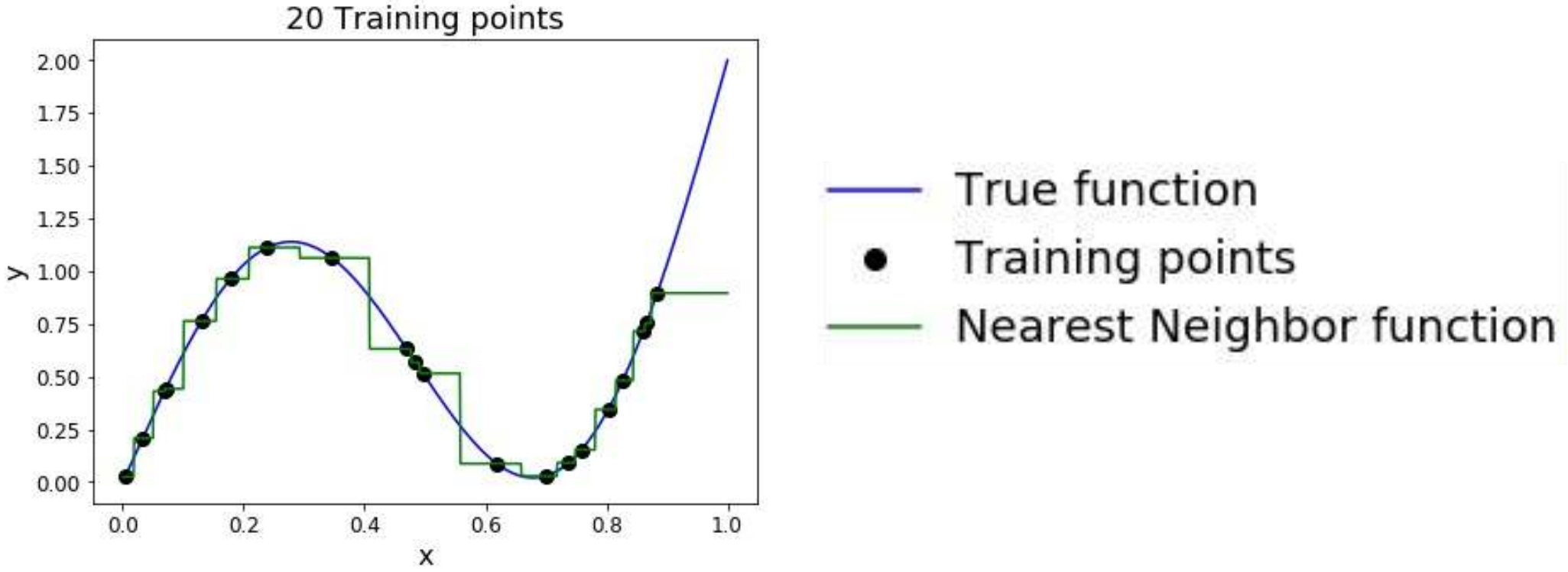


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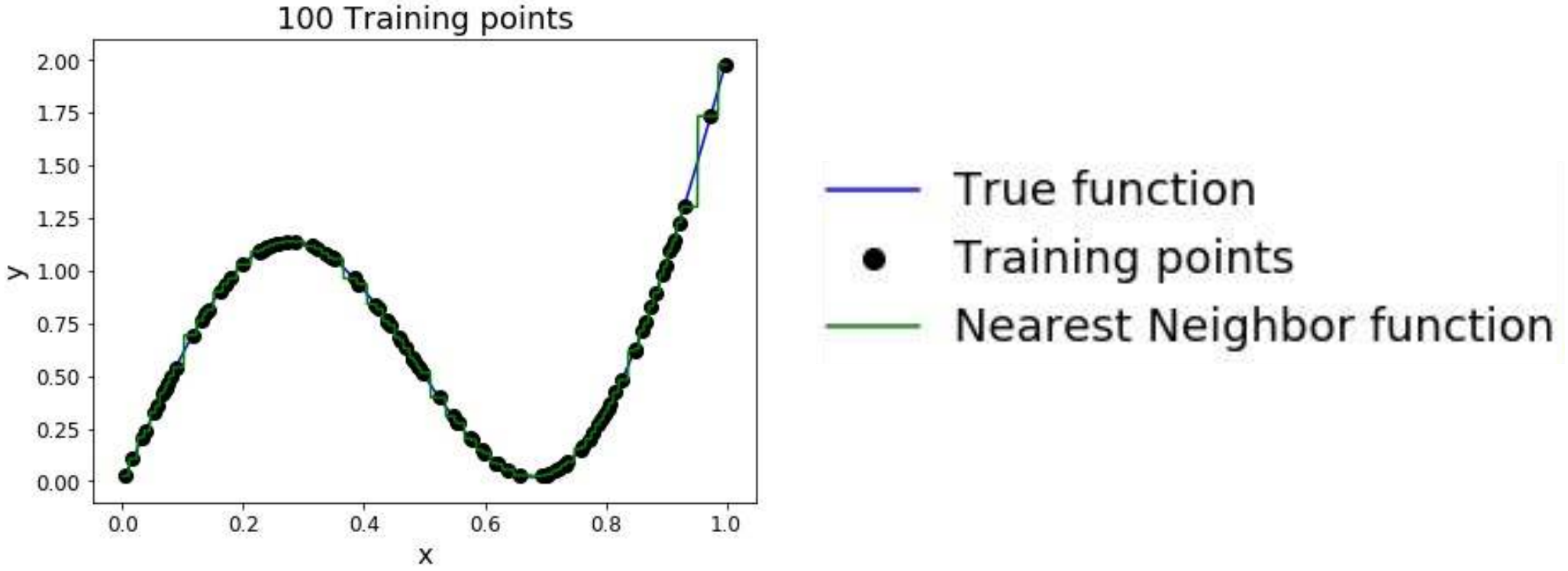


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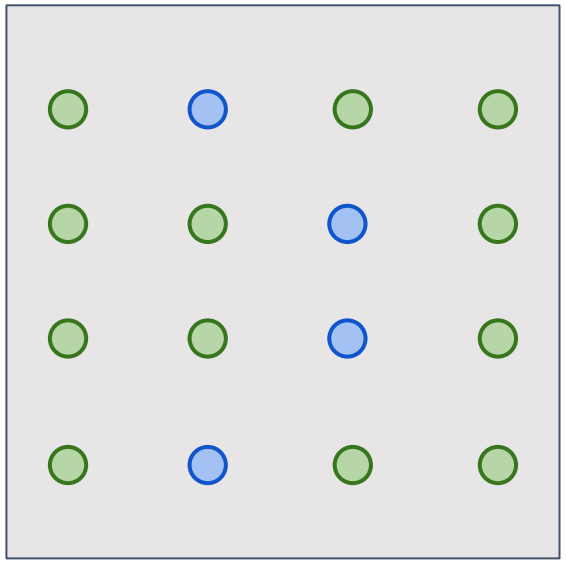
# Problem: Curse of Dimensionality

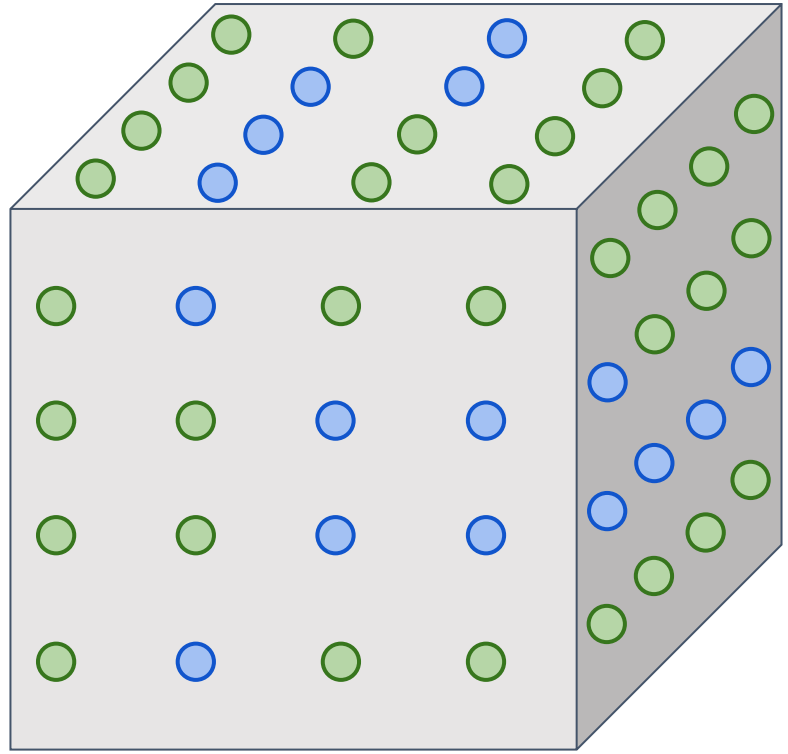
**Curse of dimensionality**: For uniform coverage of space, number of training points needed grows exponentially with dimension

Dimensions = 1
Points = 4

Dimensions = 2
Points = $4^2$



Dimensions = 3
Points = $4^3$



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Number of possible 32x32 binary images:

$$2^{32x32} \approx 10^{308}$$

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**Curse of dimensionality**: For uniform coverage of space, number of training points needed grows exponentially with dimension

Number of possible 32x32 binary images:

$2^{32x32} \approx 10^{308}$

Number of elementary particles in the visible universe: (source)

$\approx 10^{97}$

# K-Nearest Neighbor on raw pixels is **seldom used**

- Very slow at test time
- Distance metrics on pixels are not informative

Original

Boxed

Shifted

Tinted

(all 3 images have same L2 distance to the one on the left)

# Nearest Neighbor with ConvNet features works well!

Devlin et al, “Exploring Nearest Neighbor Approaches for Image Captioning”, 2015

# Nearest Neighbor with ConvNet features works well!

## Example: Image Captioning with Nearest Neighbor

A bedroom with a bed and a couch.

A cat sitting in a bathroom sink.

A train is stopped at a train station.

A wooden bench in front of a building.

Devlin et al, “Exploring Nearest Neighbor Approaches for Image Captioning”, 2015

# Summary

In **Image classification** we start with a **training set** of images and labels, and must predict labels on the **test set**

Image classification is challenging due to the semantic gap: we need invariance to occlusion, deformation, lighting, intraclass variation, etc

Image classification is a **building block** for other vision tasks

The **K-Nearest Neighbors** classifier predicts labels based on nearest training examples

Distance metric and K are **hyperparameters**

Choose hyperparameters using the **validation set**; only run on the test set once at the very end!

# Next time: Linear Classifiers